Fourth Year Pharmacy Students – Assiut University
Elective Medicinal Chemistry Course – First Semester

Drug Design and Discovery
Lecture 2

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Drug design and development

Stages:

1) Identify target disease
2) Identify drug target
3) Establish testing procedures
4) Find a lead compound
5) Structure Activity Relationships (SAR)
6) Identify a pharmacophore
7) Drug design - optimising target interactions
8) Drug design - optimising pharmacokinetic properties
9) Toxicological and safety tests
10) Chemical development and production
11) Patenting and regulatory affairs
12) Clinical trials

Optimising pharmacokinetic properties

Pharmacokinetics :

Is the study of the time course of compound and metabolite concentrations in the body.

After a compound is administered, its concentration in the bloodstream and tissues changes with time, first increasing as it enters systemic circulation and then decreasing as it is distributed to tissues, metabolized, and eliminated.

The PK parameters of a compound (e.g., clearance, half-life, volume of distribution) result from its physicochemical and biochemical properties. These properties are determined by the structure of the compound and the physical and biochemical environment into which the compound is dosed.

Aims

- To improve pharmacokinetic properties of lead compound

- To optimise chemical and metabolic stability
  (stomach acids / digestive enzymes / metabolic enzymes)

- To optimise hydrophilic / hydrophobic balance
  (solubility in blood / solubility in GIT / solubility through cell membranes / access to CNS / excretion rate)

Improving Pharmacokinetics properties

Strategies

1- Solubility and membrane permeability

2. Making drugs more resistant to chemical and enzymatic degradation

3. Making drugs less resistant to drug metabolism

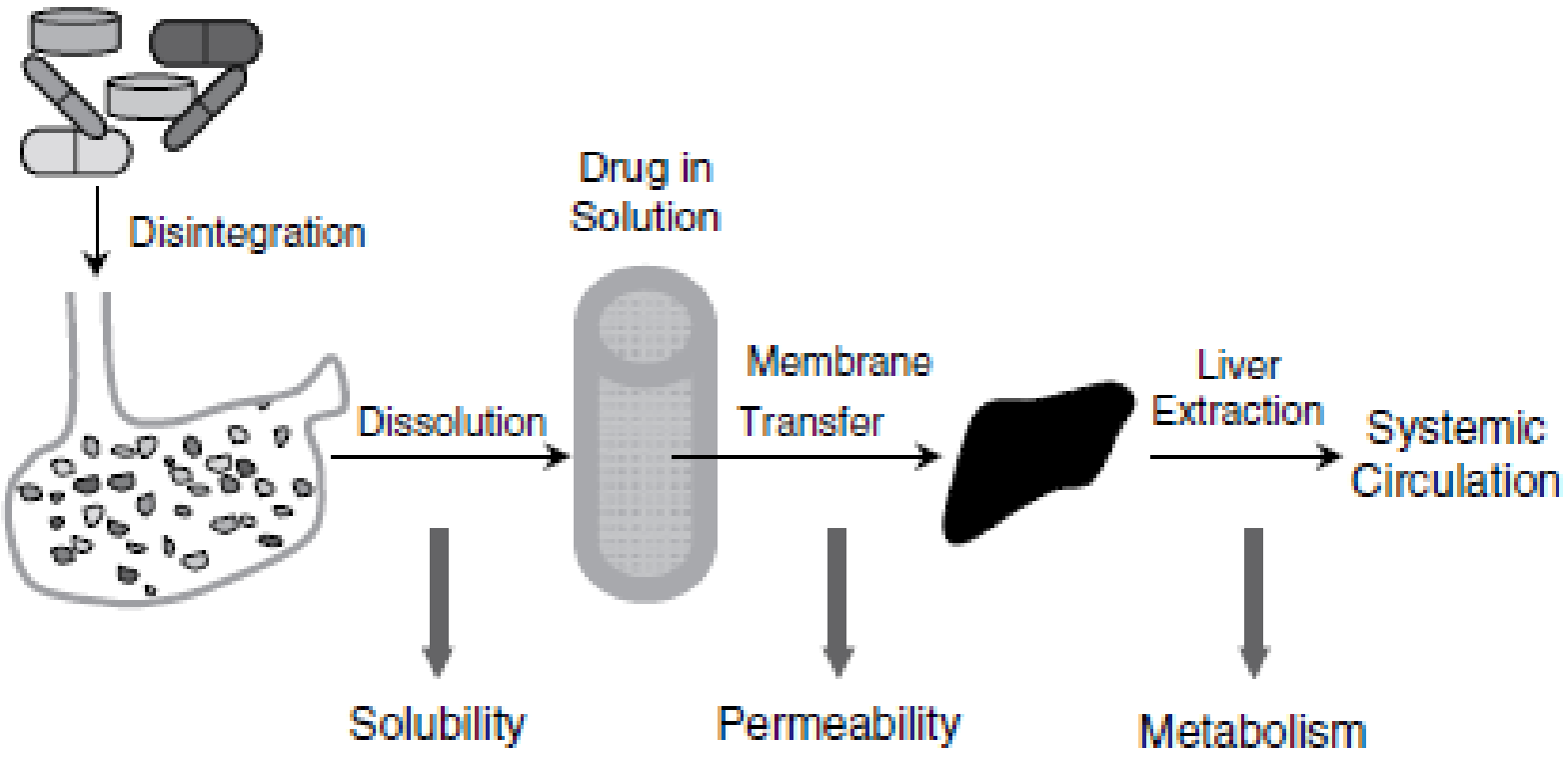
3- Drug targeting

4- Reducing drug toxicity

5- Prodrugs

6- Drug Synergy

1- Solubility and membrane permeability


Disintegration
Drug in
Solution
Dissolution
Membrane
Transfer
Liver
Extraction
Systemic
Circulation
Solubility
Permeability
Metabolism

- **<u>Drugs must be polar</u>**

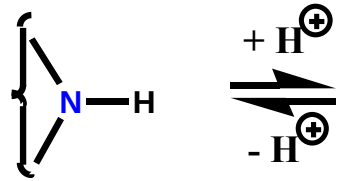
- to be soluble in aqueous conditions
- to interact with molecular targets

- **<u>Drugs must be 'fatty'</u>**

- to cross cell membranes
- to avoid rapid excretion

- Drugs must have both hydrophilic and lipophilic characteristics

- Many drugs are weak bases with pK_a's 6-8



Crosses membranes

Receptor interactior & water solubility

1. Solubility and membrane permeability

1.1 Vary polarity

Variation of alkyl or acyl substituents

Adding polar groups

Masking or removing polar groups

Bioisosters for polar groups

1.2 Vary pK_a

Variation of N-alkyl substituent

Variation of aromatic substituent

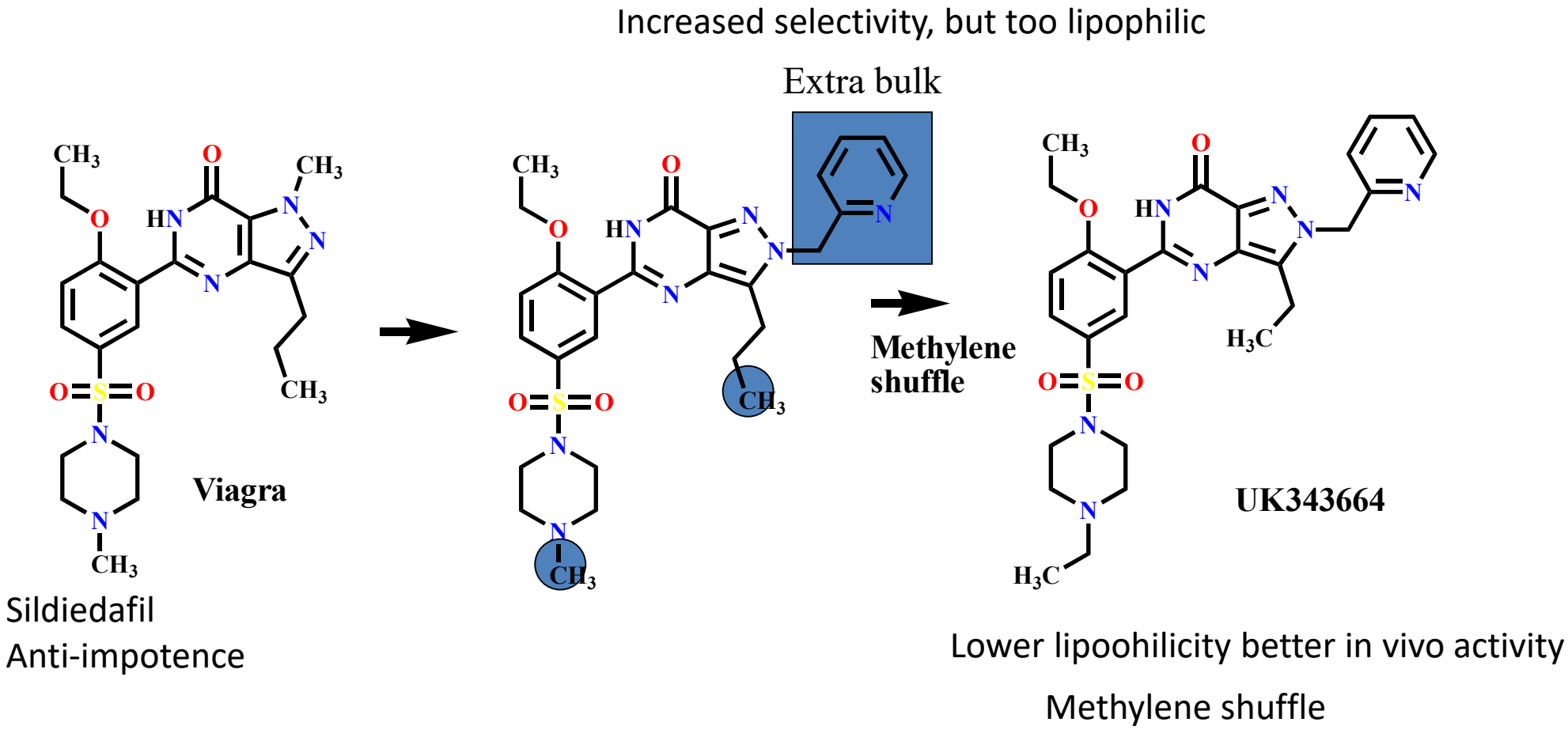
1.1 Vary polarity

Variation of alkyl or acyl substituents

- Varying the size of alkyl groups varies the hydrophilic / hydrophobic balance of the structure
- Larger alkyl groups increase hydrophobicity

Disadvantage:

- May interfere with target binding for steric reasons



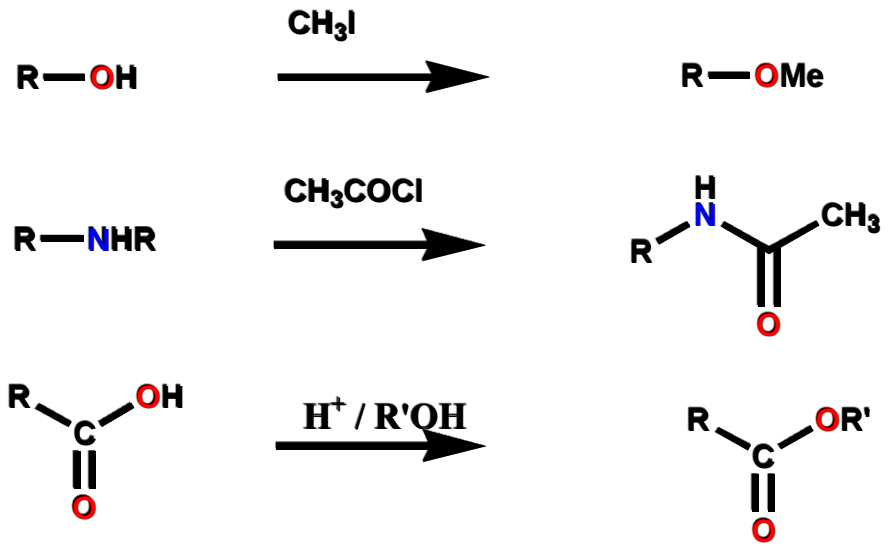
methylene shuffle: increasing the size of one alkyl group and decreasing the size of another has been found to modify the hydrophobicity of a compound

‘Masking’ or removing polar groups

Masking or removing polar groups decreases polarity and increases hydrophobic character

Disadvantages:

- Polar group may be involved in target binding
- Unnecessary polar groups are likely to have been removed already (simplification strategy)



Adding polar groups

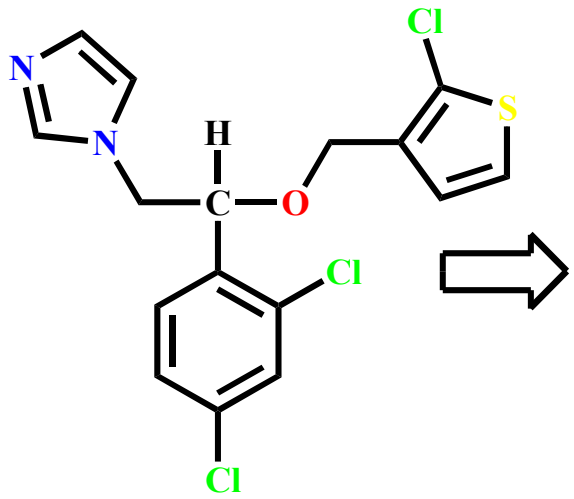
Adding polar groups increases polarity and decreases hydrophobic character.

Useful for targeting drugs vs. gut infections.

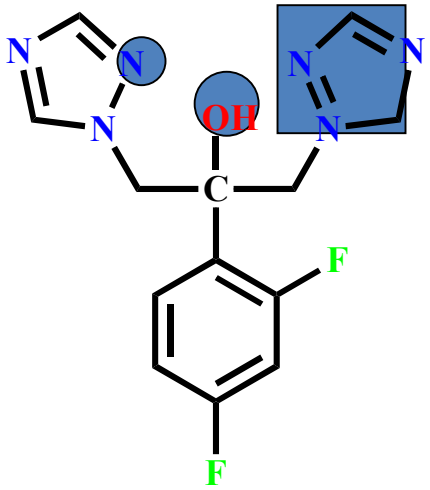
Useful for reducing CNS side effects.

Disadvantage:

May introduce unwanted side effects



Tioconazole



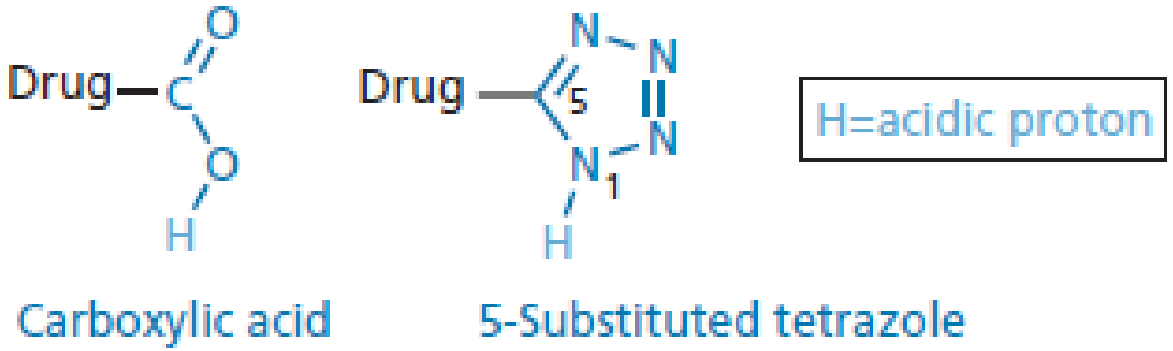
Fluconazole

Antifungal agent with poor solubility - skin infections only

Systemic antifungal agent improved blood solubility

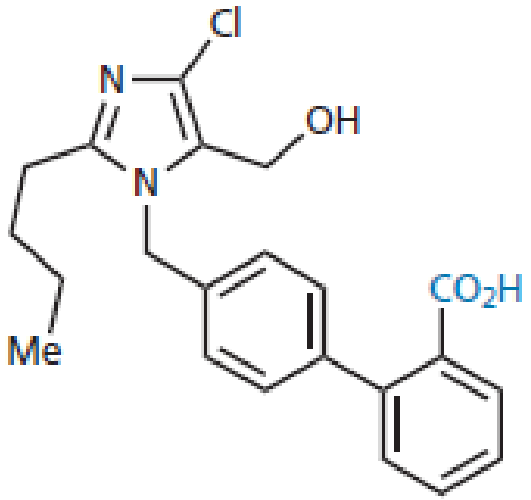
Bioisosters for polar groups

For example, a,carboxylic acid is a highly polar group which can ionize and hinder absorption of any drug containing it.

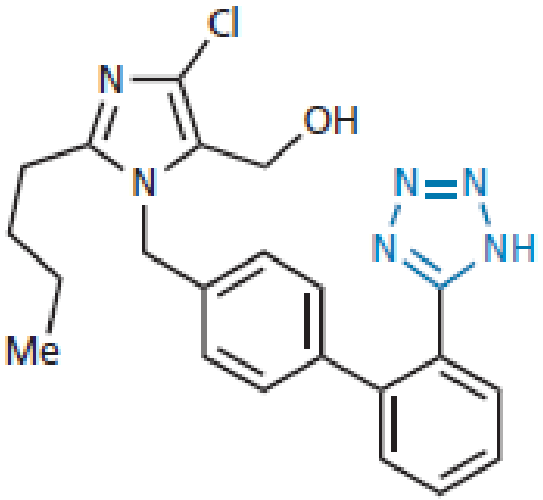


Advantages:

- Like carboxylic acids, tetrazoles contain an acidic proton and are ionized at pH 7.4 and planer.
- tetrazole anion is 10 times more lipophilic than a carboxylate anion and drug absorption is enhanced .
- Resistant to many of the metabolic reactions that occur on carboxylic acids



Structure I



Losartan

FIGURE 1 Development of losartan.

Angiotensin II inhibitor given IV

1.2 Vary pK_a

Overview

- The ionizability of a compound is indicated by pKa

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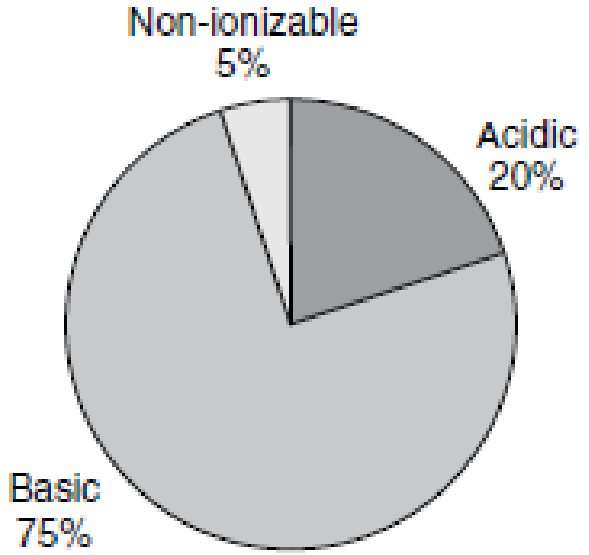
- Ionizability is a major determinant of solubility and permeability.

- When pH=pKa, the concentrations of ionized and neutral molecules in solution are equal.

- Basicity of bases increases as pKa increases; acidity of acids increases at pKa decreases.

- Varying pK_a alters percentage of drug which is ionised alter pK_a to obtain require ratio of ionized to unionized drug

- Drugs with a p *K a outside the range 6–9 tend to be too* strongly ionized and are poorly absorbed through cell membranes .



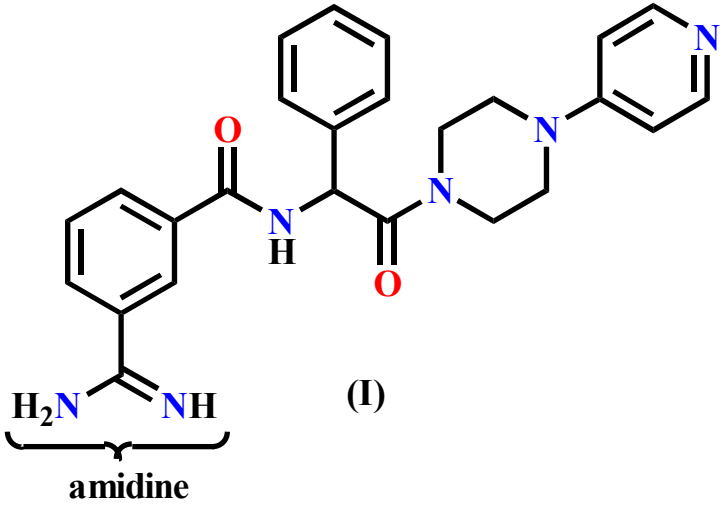
Disadvantage:

- May affect binding interactions

Variation of N-alkyl substituents to vary pK a

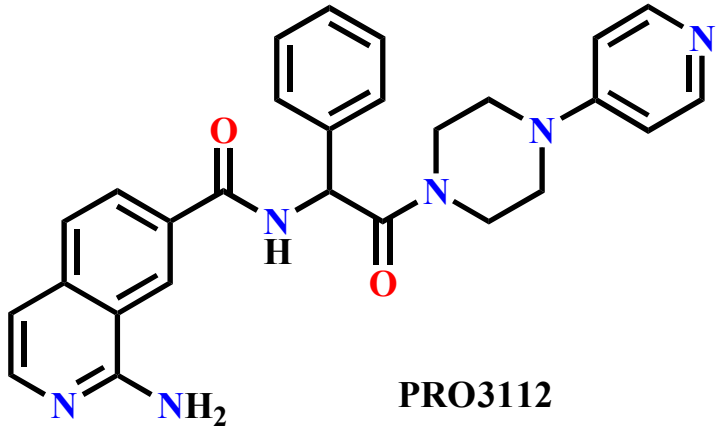
Extra *N –alkyl* groups or larger *N -alkyl groups have an increased electron*- donating eff ect which should increase basicity

But increasing the size or number of alkyl groups increases the steric bulk around the nitrogen atom. This hinders water molecules from solvating the ionized form of the base and prevents stabilization of the ion. This, in turn, decreases the basicity of the amine.



(I)

Antithrombotic
but too basic



PRO3112

Decreased basicity

N locked into heterocycle isoquinoline ring

2. Making drugs more resistant to chemical and enzymatic degradation

Strategies:

2.1 Steric Shields

2.2 ‘Electronic shielding’ of NH_2

2.3 Stereoelectronic Effects

2.4 Metabolic blockers

2.5 Remove / replace susceptible metabolic groups

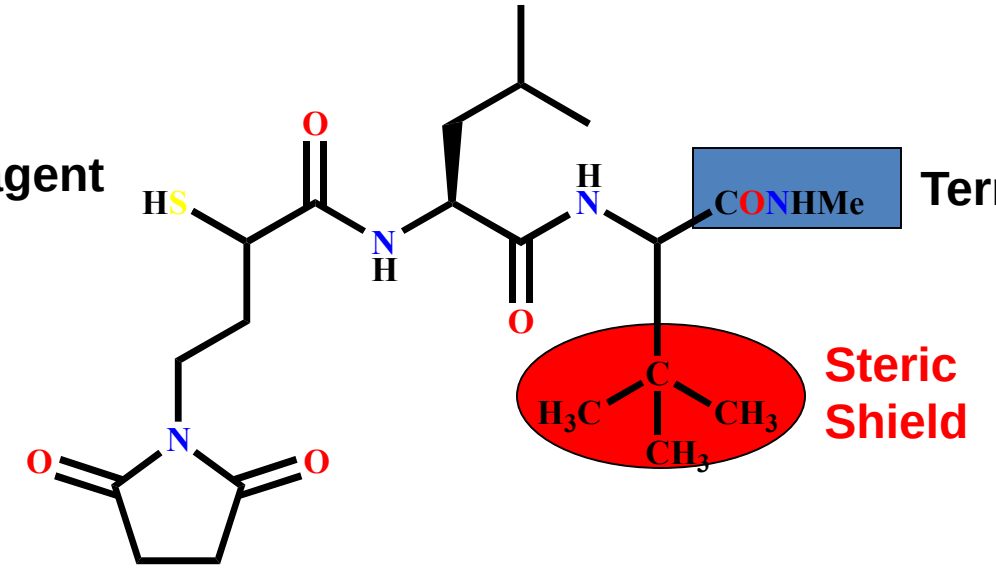
2.6 Shifting susceptible metabolic groups

2.7 Ring variation and ring substituents

2.1 Steric Shields

- Used to increase chemical and metabolic stability
- Introduce bulky group as a shield
- Protects a susceptible functional group from hydrolysis
- Hinders attack by nucleophiles or enzymes

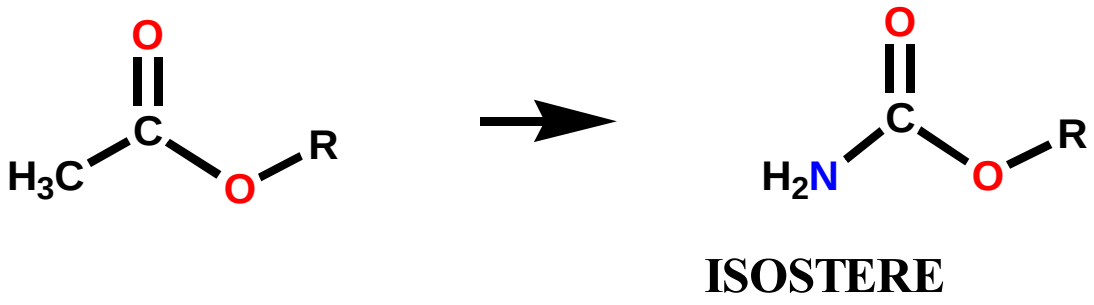
Antirheumatic agent
D1927



Blocks hydrolysis of terminal amide

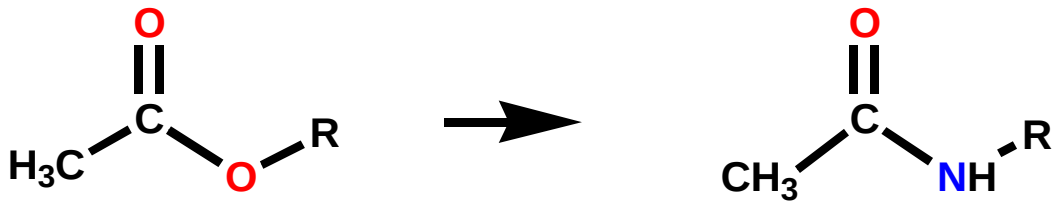
2.2 'Electronic shielding' of NH_2

- Used to stabilise labile functional groups (e.g. esters)
- Replace labile ester with more stable urethane or amide
- Nitrogen feeds electrons into carbonyl group and makes it less reactive
- Increases chemical and metabolic stability



urethane

ISOSTERE



amide group

ISOSTERE

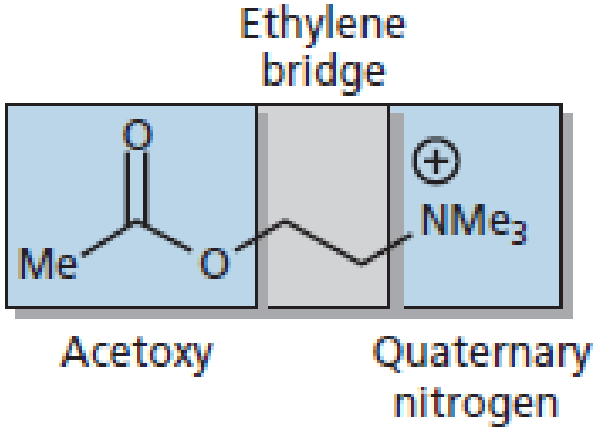


FIGURE 22.9 Acetylcholine.

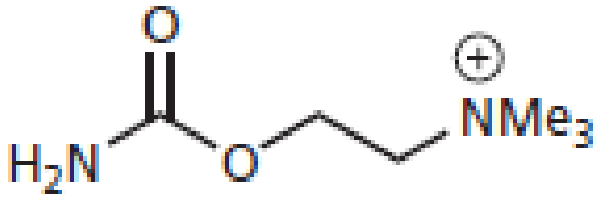


FIGURE 22.19 Carbachol.

long-acting cholinergic agent

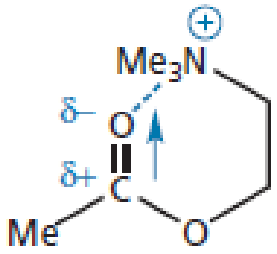


FIGURE 22.16 Neighbouring group participation. The arrow indicates the inductive pull of oxygen which increases the electrophilicity of the carbonyl carbon (see Molecular modelling exercise 22.1).

neighbouring group participation

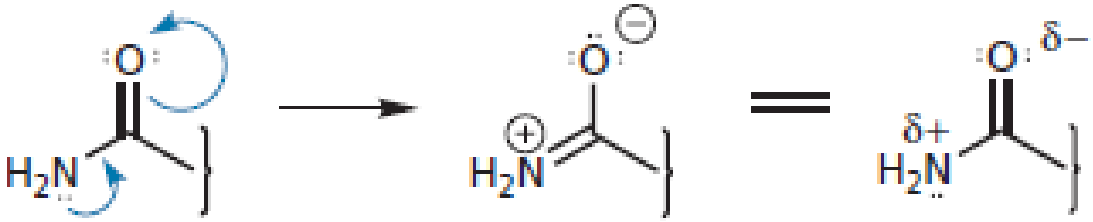
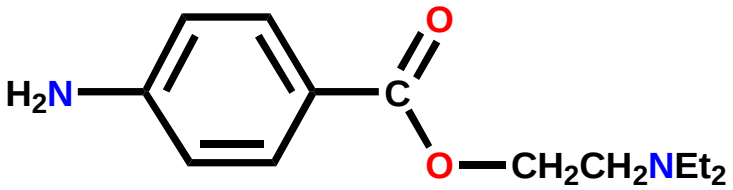


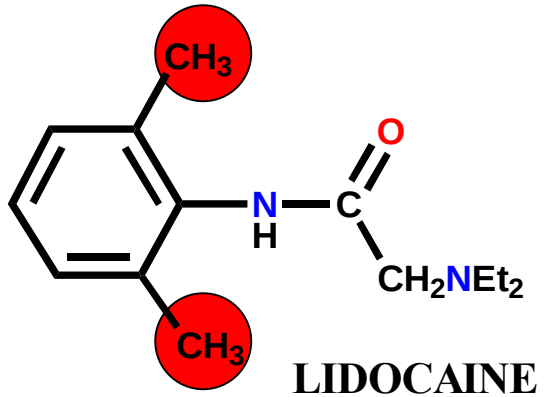
FIGURE 22.20 Resonance structures of carbachol.

2.3 Stereoelectronic Effects

- Steric and electronic effects used in combination
- Increases chemical and metabolic stability



PROCAINE

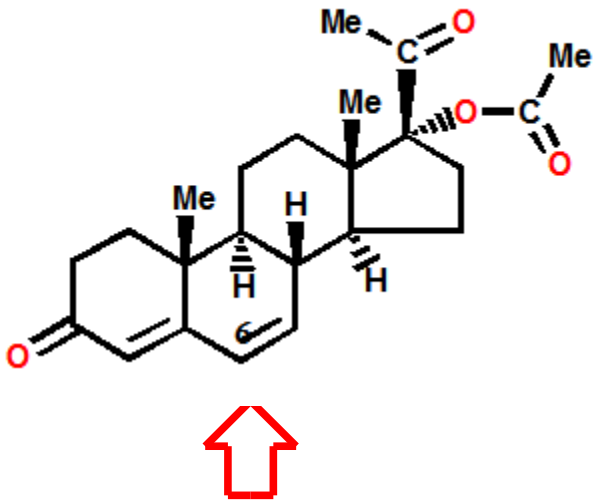


LIDOCAINE

***Ortho* Methyl groups act as steric shields & hinder hydrolysis by esterases**
Amide more stable than ester (electronic effect)

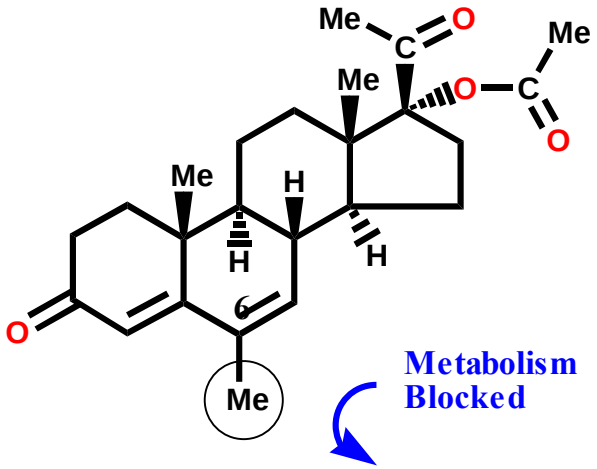
2.4 Metabolic blockers

- Metabolism of drugs usually occur at specific sites. Introduce groups at a susceptible site to block the reaction
- Increases metabolic stability and drug lifetime



Metabolic Oxidation

Oral contraceptive
- limited lifetime



Megestrol acetate

Replacing a hydrogen atom with a deuterium isotope.

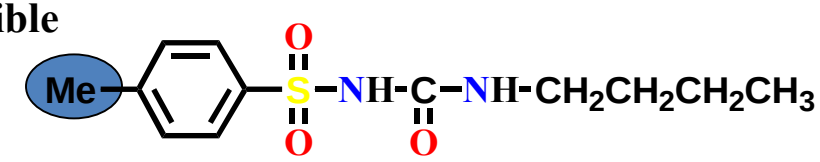
- The covalent bond between carbon and deuterium is twice as strong as that between carbon and hydrogen.

- Help to block metabolic mechanisms.

2.5 Remove / replace susceptible metabolic groups

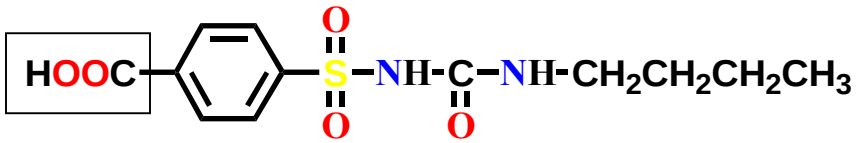
- Metabolism of drugs usually occurs at specific groups.
- Remove susceptible group or replace it with metabolically stable group

Susceptible group



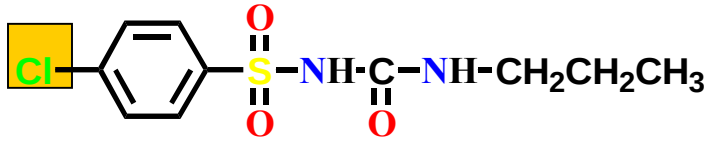
TOLBUTAMIDE

Metabolism ↓

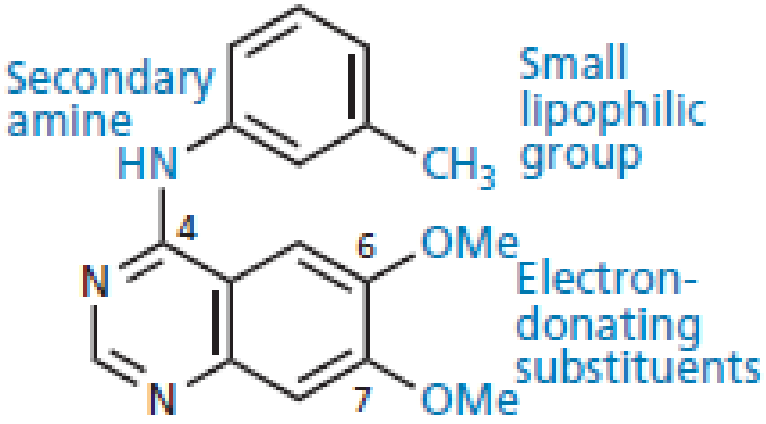


Rapidly excreted - short lifetime

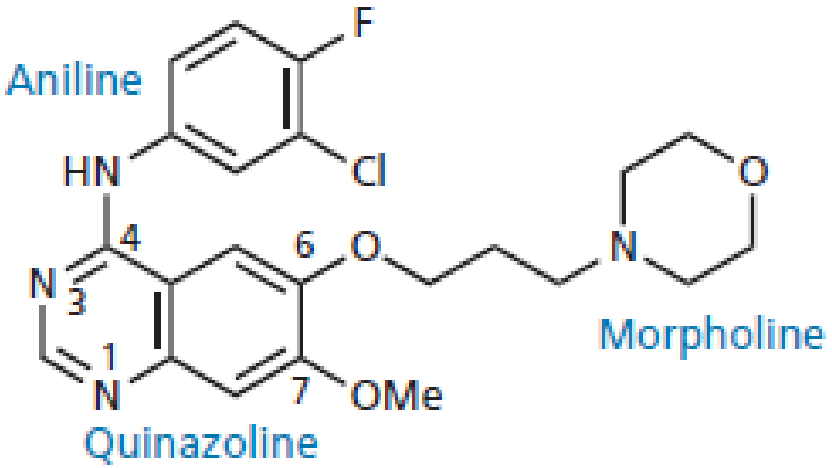
Unsusceptible group
chlorpropamide



Metabolism ⇓



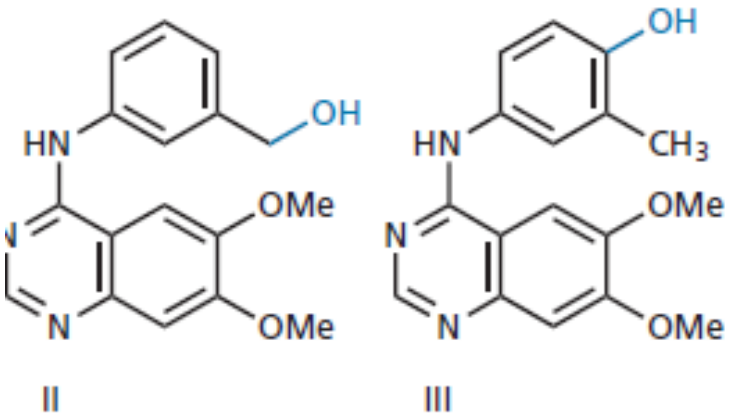
I (IC_{50} 5 nM)



(anticancer)EGF-R kinase inhibitors

Stable analouge

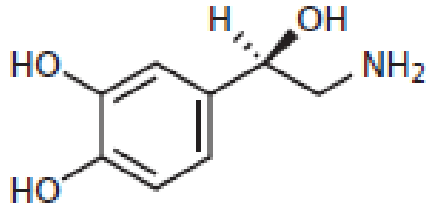
Geftinib



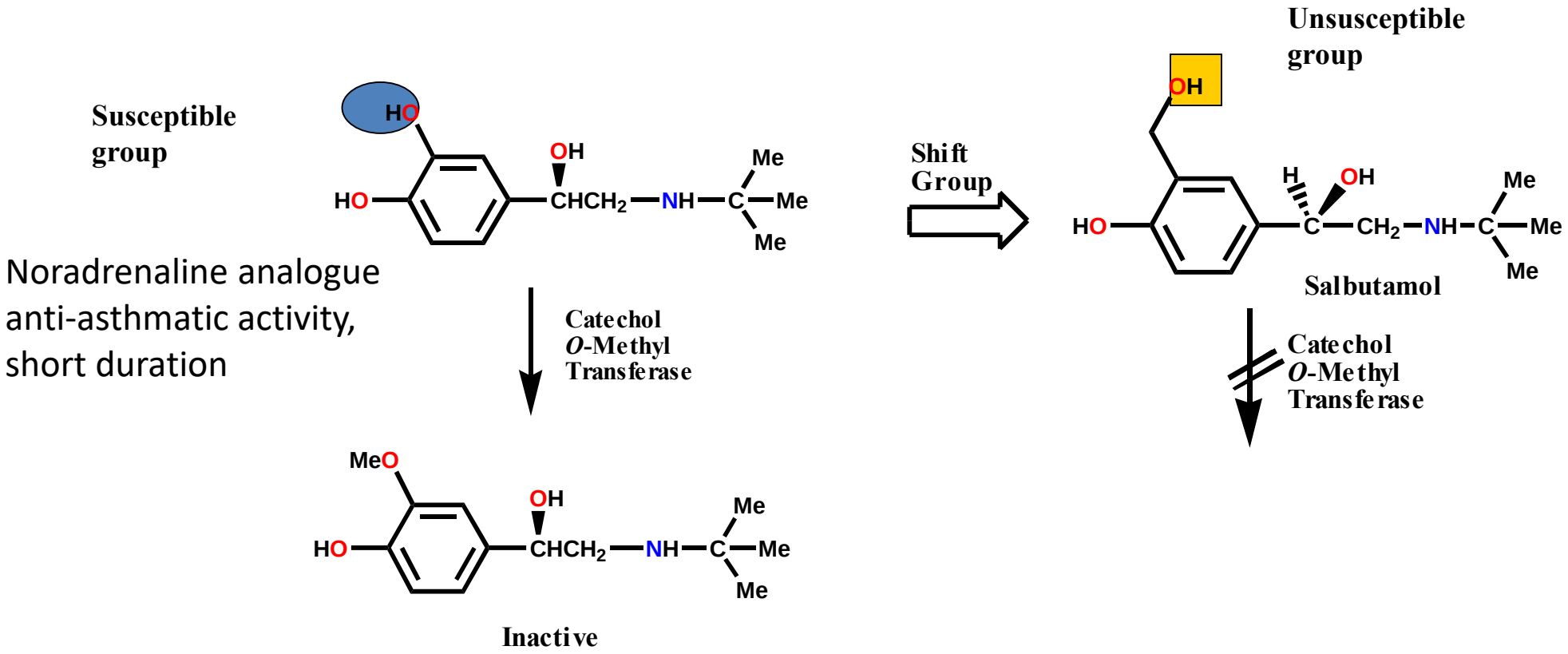
Metabolites

2.6 Shifting susceptible metabolic groups

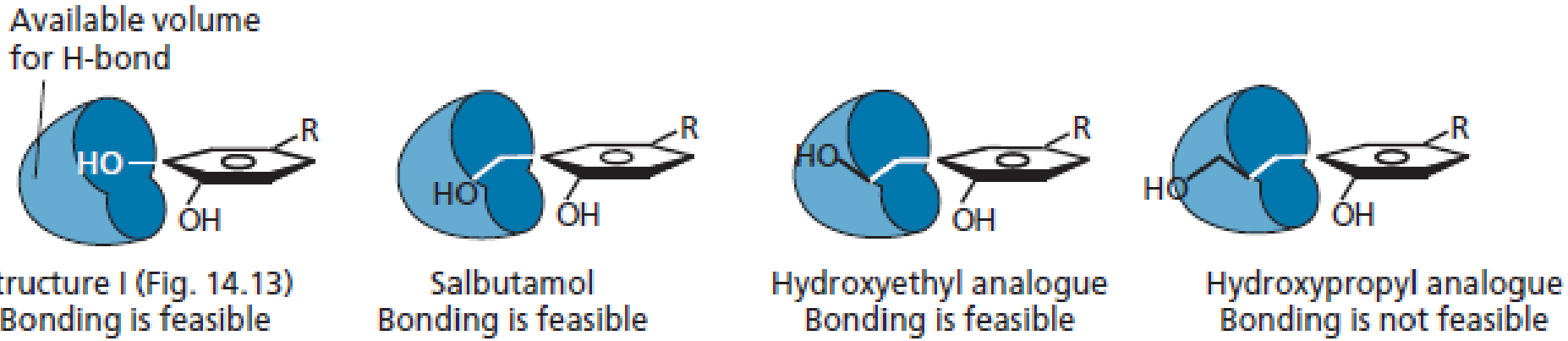
- Used if the metabolically susceptible group is important for binding
- Shift its position to make it unrecognisable to metabolic enzyme
- Must still be recognisable to target



Noradrenaline



Shifting an important binding group that is metabolically susceptible cannot be guaranteed to work in every situation.
It may well make the molecule unrecognizable both to its target and to the metabolic enzyme.



Structure I (Fig. 14.13)
Bonding is feasible

Salbutamol
Bonding is feasible

Hydroxyethyl analogue
Bonding is feasible

Hydroxypropyl analogue
Bonding is not feasible

2.7 Ring variation and ring substituents

Essential for activity

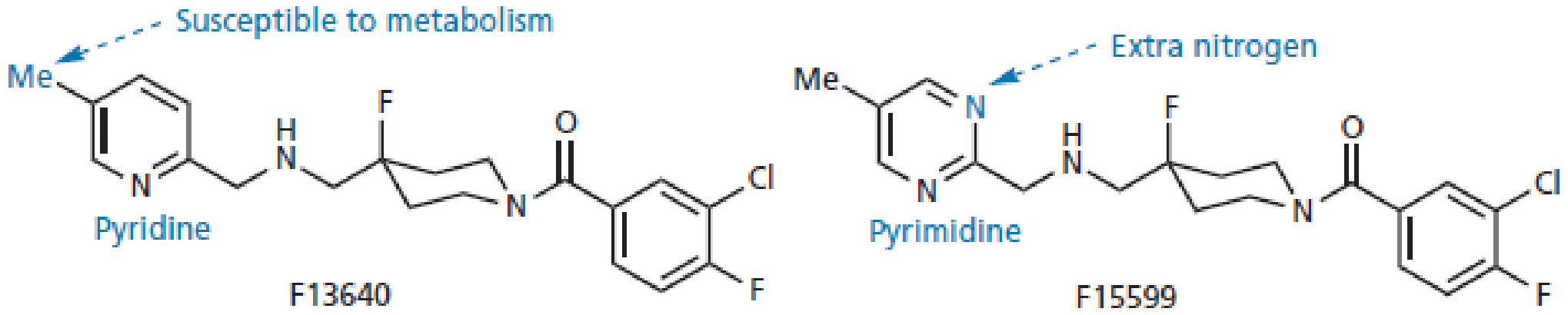


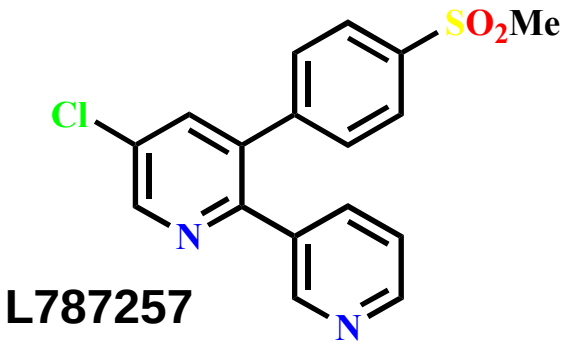
FIGURE 14.14 Stabilizing an aromatic or heteroaromatic methyl substituent by adding a nitrogen to the ring.

3. Making drugs less resistant to drug metabolism

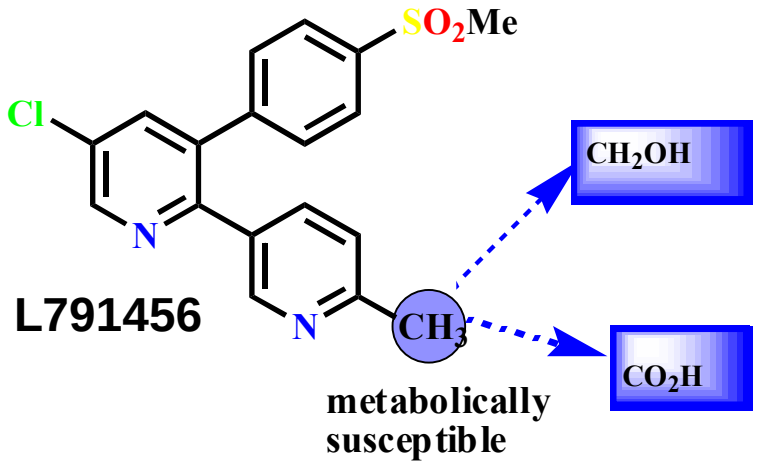
3.1 Introducing susceptible metabolic groups

- Used to decrease metabolic stability and drug lifetime
- Used for drugs which ‘linger’ too long in the body and cause side effects
- Add groups known to be susceptible to Phase I or Phase II metabolic reactions

Anti-arthritic agents

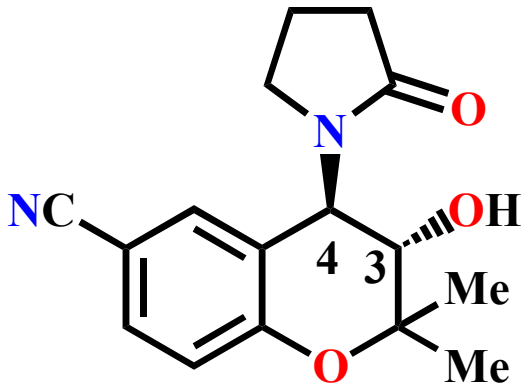


L787257



L791456

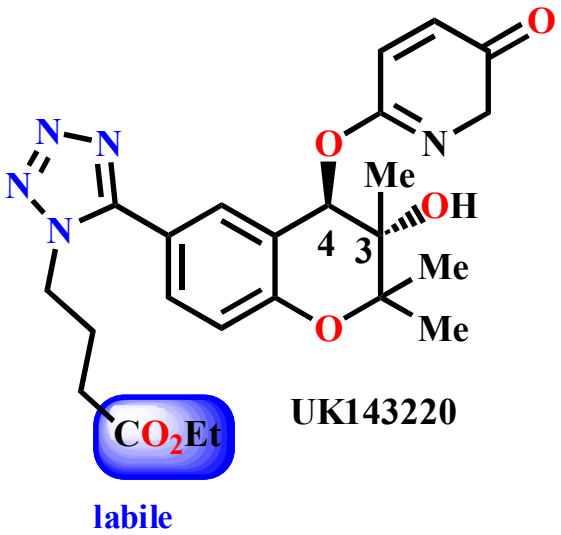
metabolically susceptible



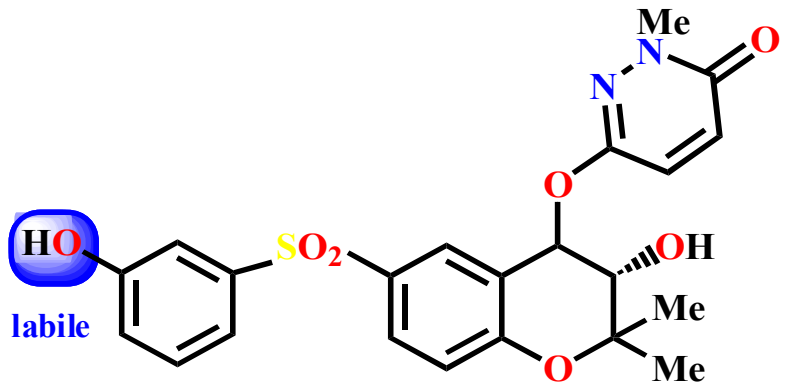
Cromakalim

Anti-asthmatic agents

Causes cardiovascular side effects



UK143220



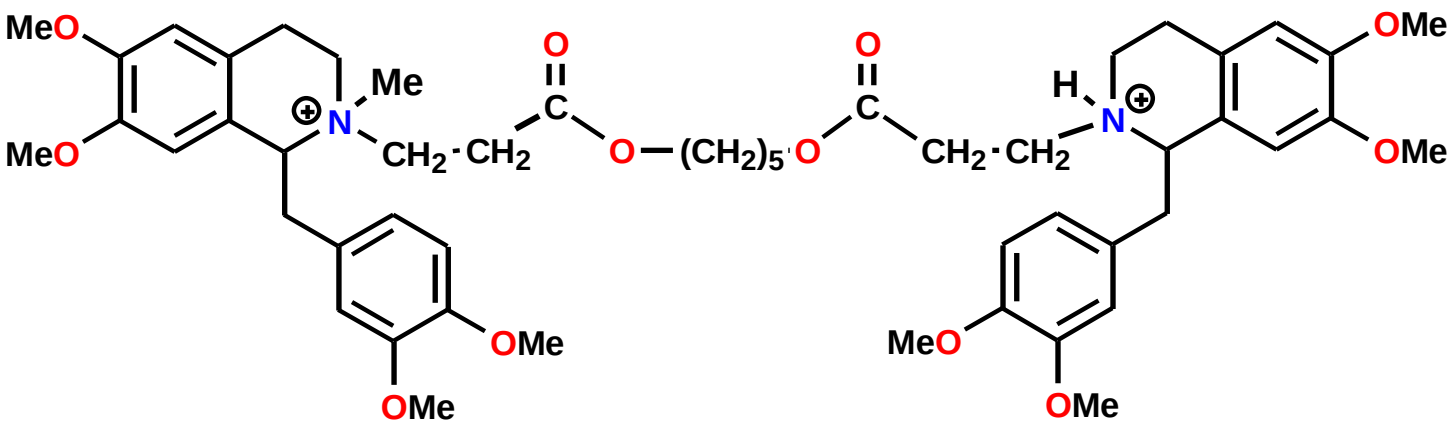
UK157147

- Cromakalim produces cardiovascular side effects if it reaches blood supply
- Add metabolic instability such that compound rapidly metabolised in blood
- UK143220 - ester quickly hydrolysed by esterases to inactive acid
- UK 157147- phenol quickly conjugated and eliminated

3.2 Self-destruct drugs

A **self-destruct drug** is one which is chemically stable under one set of conditions, but becomes unstable and degrades spontaneously under another set of conditions.

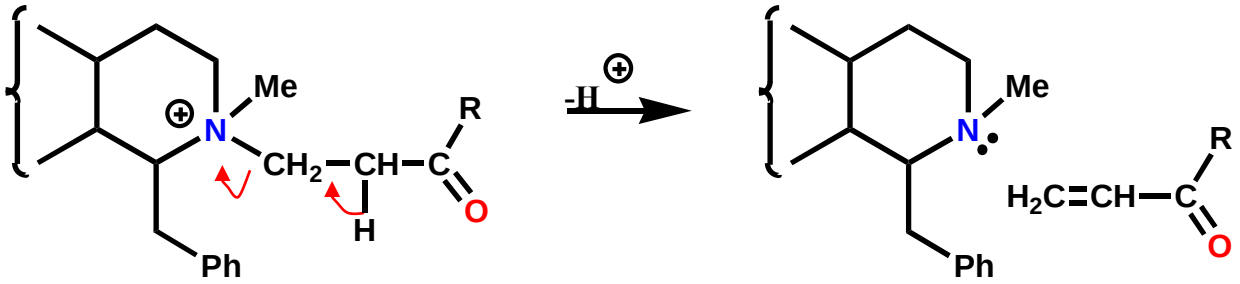
The advantage of a self-destruct drug is that inactivation does not depend on the activity of metabolic enzymes,



Atracurium - i.v. neuromuscular blocking agent

- Stable at acid pH, unstable at blood pH (slightly alkaline)
- Self destructs by Hoffmann elimination and has short lifetime
- Anaesthetist to control dose levels accurately during surgery by providing it as a continuous, intravenous drip.
- Quick recovery times after surgery

Hoffmann Elimination



ACTIVE

INACTIVE

4- Drug targeting

4. Drug targeting

4.1 Linking a biosynthetic building block

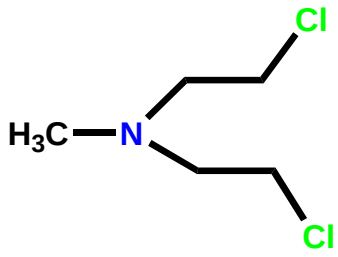
4.2 Linking drugs to monoclonal antibodies

4.3 Targeting gut infections

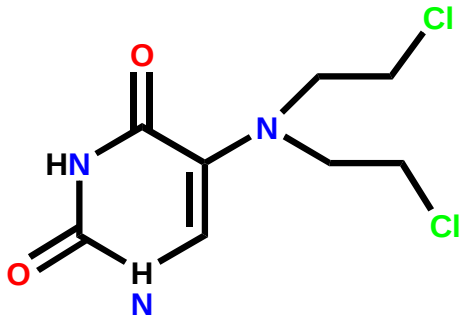
4.4 Targeting peripheral regions over CNS

4.1 Linking a biosynthetic building block

- Drug ‘smuggled’ into cell by carrier proteins for natural building block (e.g. amino acids or nucleic acid bases)
- Increases selectivity of drugs to target cells and reduces toxicity to other cells



Non selective alkylating agent
Toxic



Uracil Mustard
Anticancer drugs

- Alkylating group is attached to a nucleic acid base
- Cancer cells grow faster than normal cells and have a greater demand for nucleic acid bases
- Drug is concentrated in cancer cells - Trojan horse tactic

4.2 Linking drugs to monoclonal antibodies

Example:

Anticancer agents

Rationale:

- Identify an antigen which is overexpressed on a cancer cell
- Clone a monoclonal antibody for the antigen
- Attach a drug or poison (e.g. ricin) to the monoclonal antibody
- Antibody carries the drug to the cancer cell
- Drug is released at the cancer cell

4.3 Targeting gut infections

Rationale:

- Design the antibacterial agent to be highly polar or ionised
- Agent will be too polar to cross the gut wall
- Agent will be concentrated at the site of infection
- Example - highly ionised sulfonamides

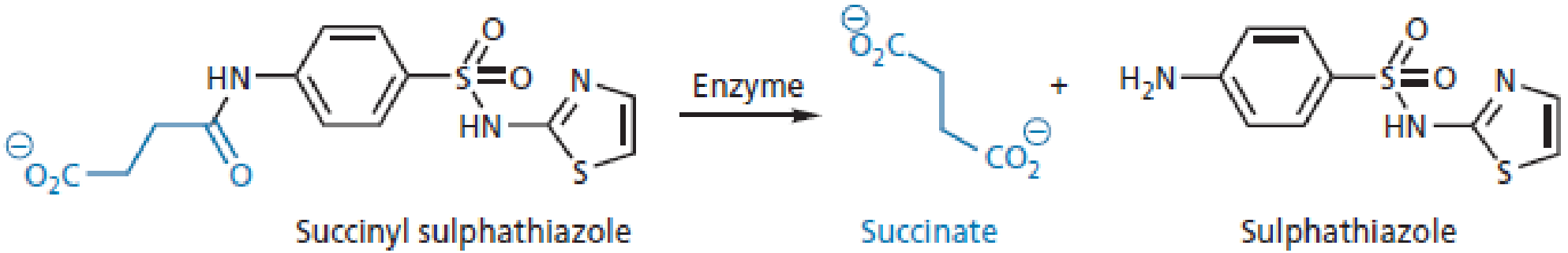


FIGURE 1 Succinyl sulphathiazole is a prodrug of sulphathiazole.

4.4 Targeting peripheral regions over CNS

Rationale:

- Increase polarity of the drug
- Drug is less likely to cross the blood brain barrier

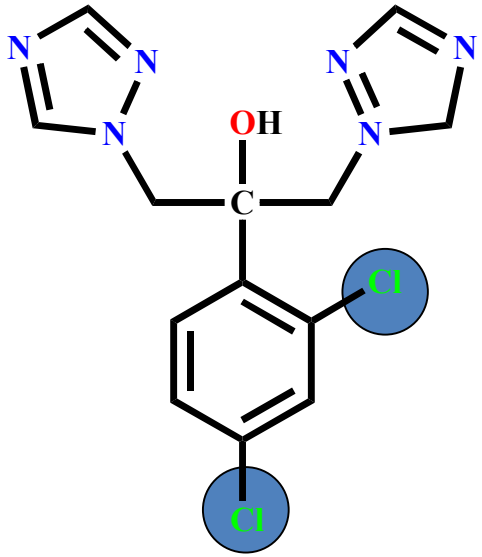
5.Reducing drug toxicity

Rationale:

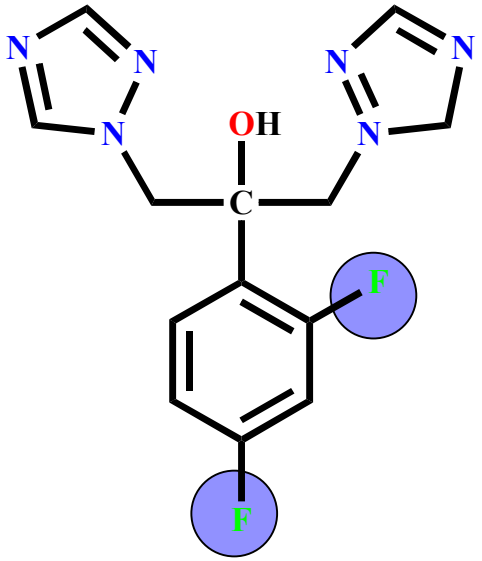
- Toxicity is often due to specific functional groups
- Remove or replace functional groups known to be toxic e.g.
  - aromatic nitro groups
  - aromatic amines
  - bromoarenes
  - hydrazines
  - polyhalogenated groups
  - hydroxylamines
- Vary substituents
- Vary position of substituents

5.1 varying substituents

- **Fluconazole (Diflucan) - antifungal agent**



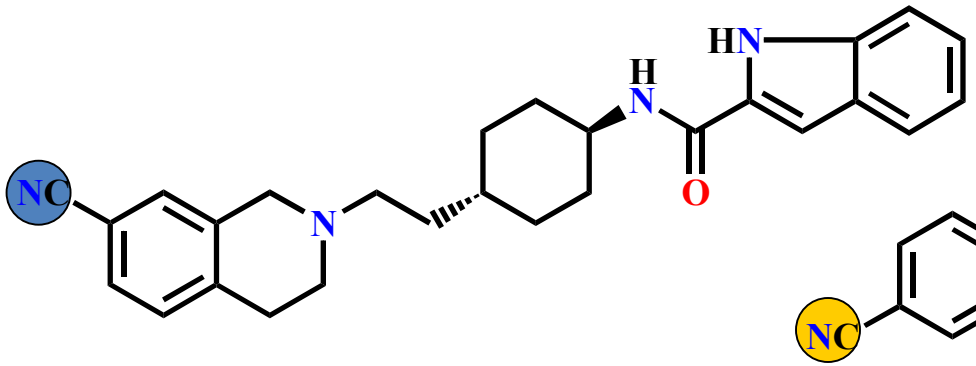
UK-47265



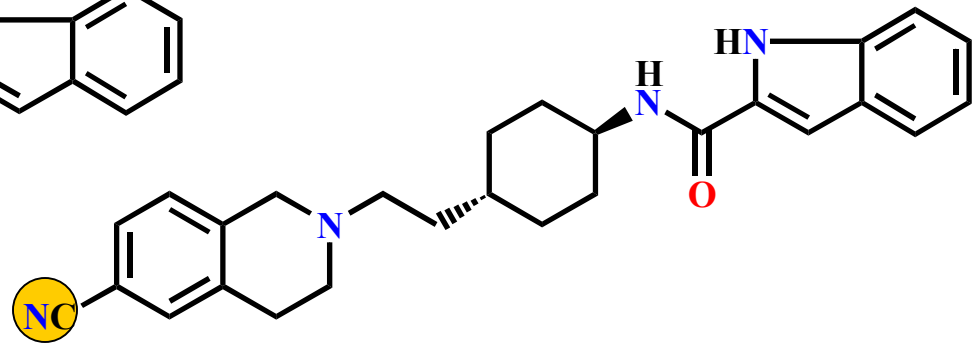
Fluconazole

Substituents varied
Less toxic

5.2 varying substituent position



Inhibits P450 enzymes



No inhibition of P450 enzymes

- Dopamine antagonists

6. Prodrugs

Definition:

Inactive compounds which are converted to active compounds in the body.

Uses:

6.1 Improving membrane permeability

6.2Prolonging activity

6.3Masking toxicity and side effects

6.4Varying water solubility

6.5Drug targeting

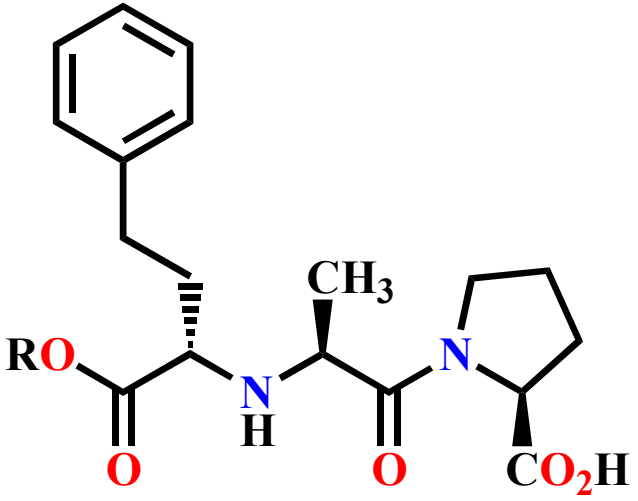
6.6Improving chemical stability

6.7‘Sleeping agents’

5.1 Prodrugs to improve membrane permeability

Esters

- Used to mask polar and ionisable carboxylic acids
- Hydrolysed in blood by esterases
- Used when a carboxylic acid is required for target binding
- Leaving group (alcohol) should ideally be non toxic



R=Et Enalapril
R=H Enalaprilit

Enalapril for enalaprilate (antihypertensive)

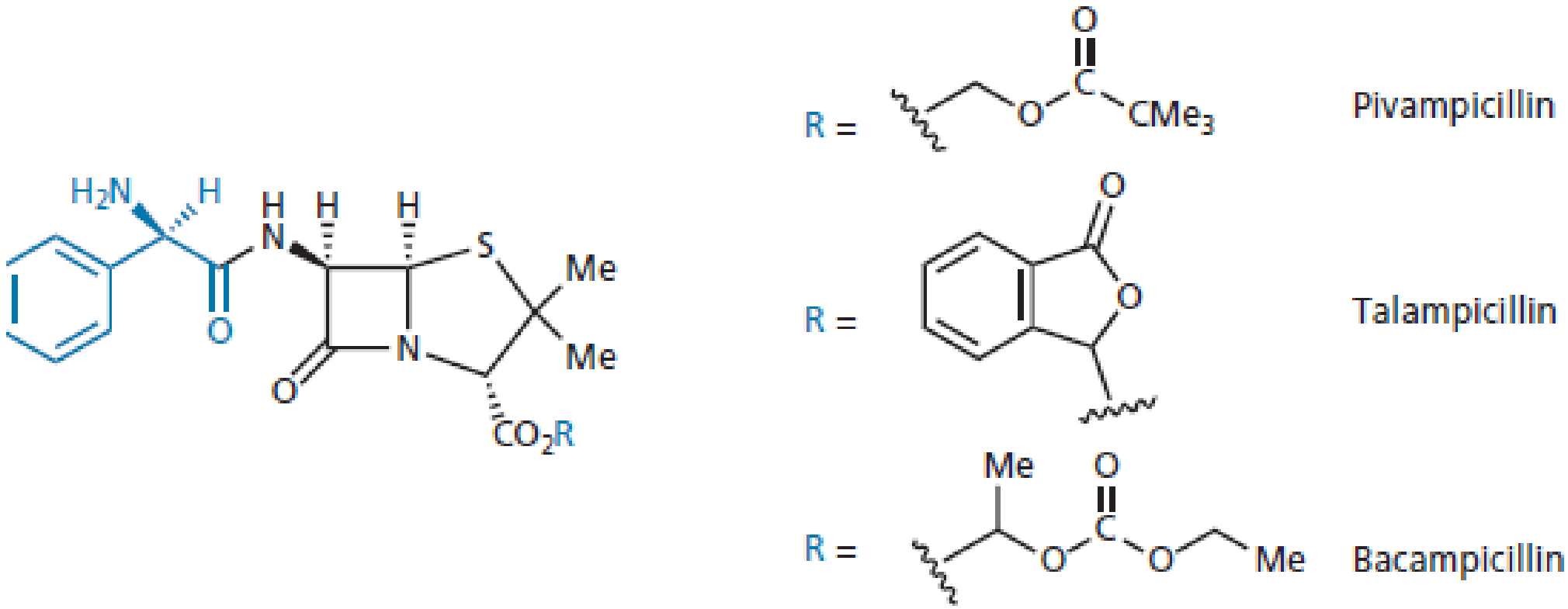
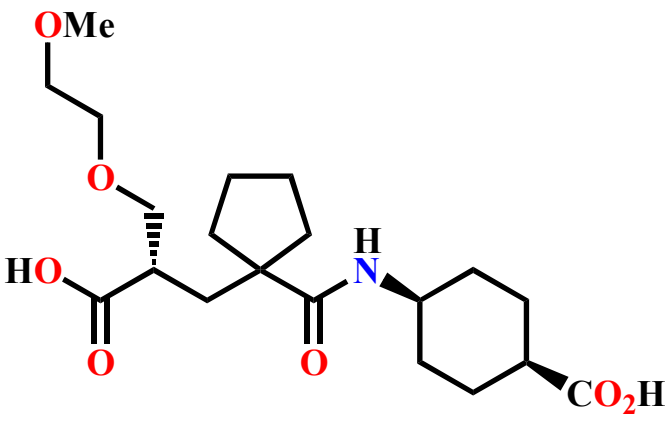
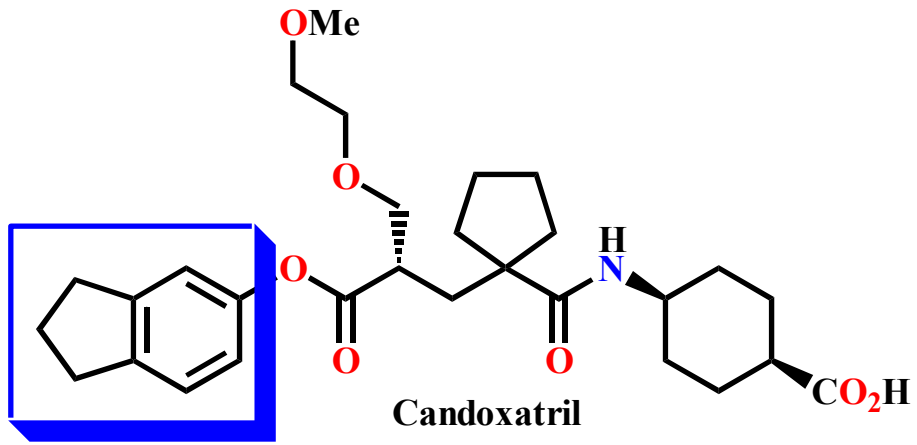


FIGURE 1 Prodrugs used to aid absorption of ampicillin through the gut wall.



Candoxatrilat

protease inhibitor



Candoxatril

5-indanyl group

- Varying the ester varies the rate of hydrolysis
- Electron withdrawing groups increase rate of hydrolysis
  (e.g. 5-indanyl)
- Leaving group (5-indanol) is non toxic

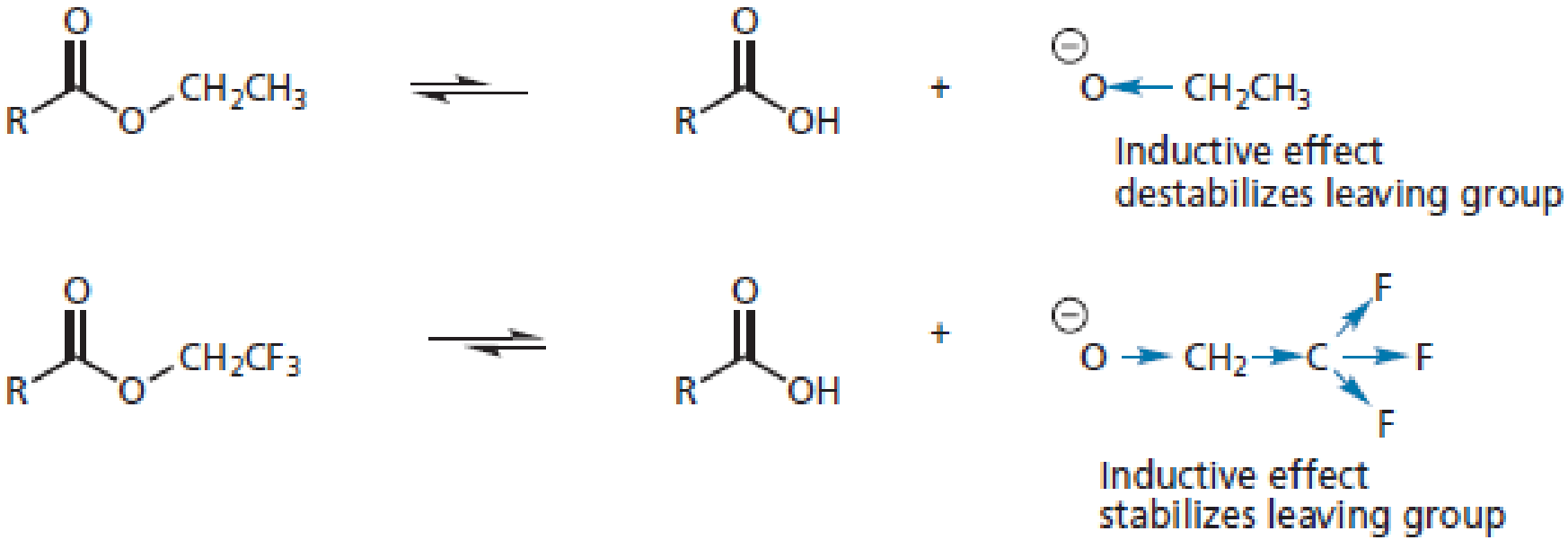


FIGURE 14.20 Inductive effects on the stability of leaving groups.

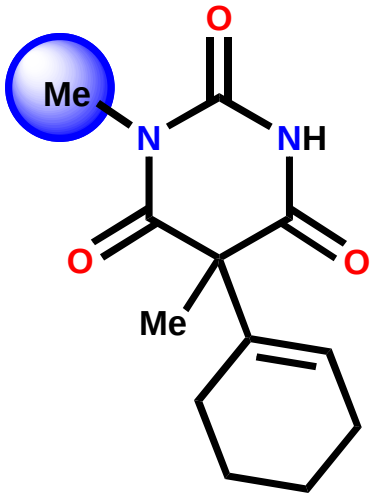
N-Methylation of amines

- **Used to reduce polarity of amines**
- **Demethylated in liver**

Example:

Hexobarbitone

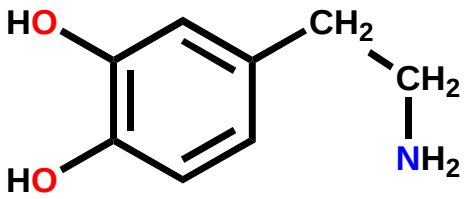
hypnotics and anti-epileptics



Trojan Horse Strategy

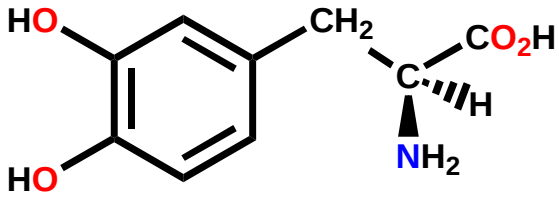
- Prodrug designed to mimic biosynthetic building block
- Transported across cell membranes by carrier proteins

Example: Levodopa for dopamine



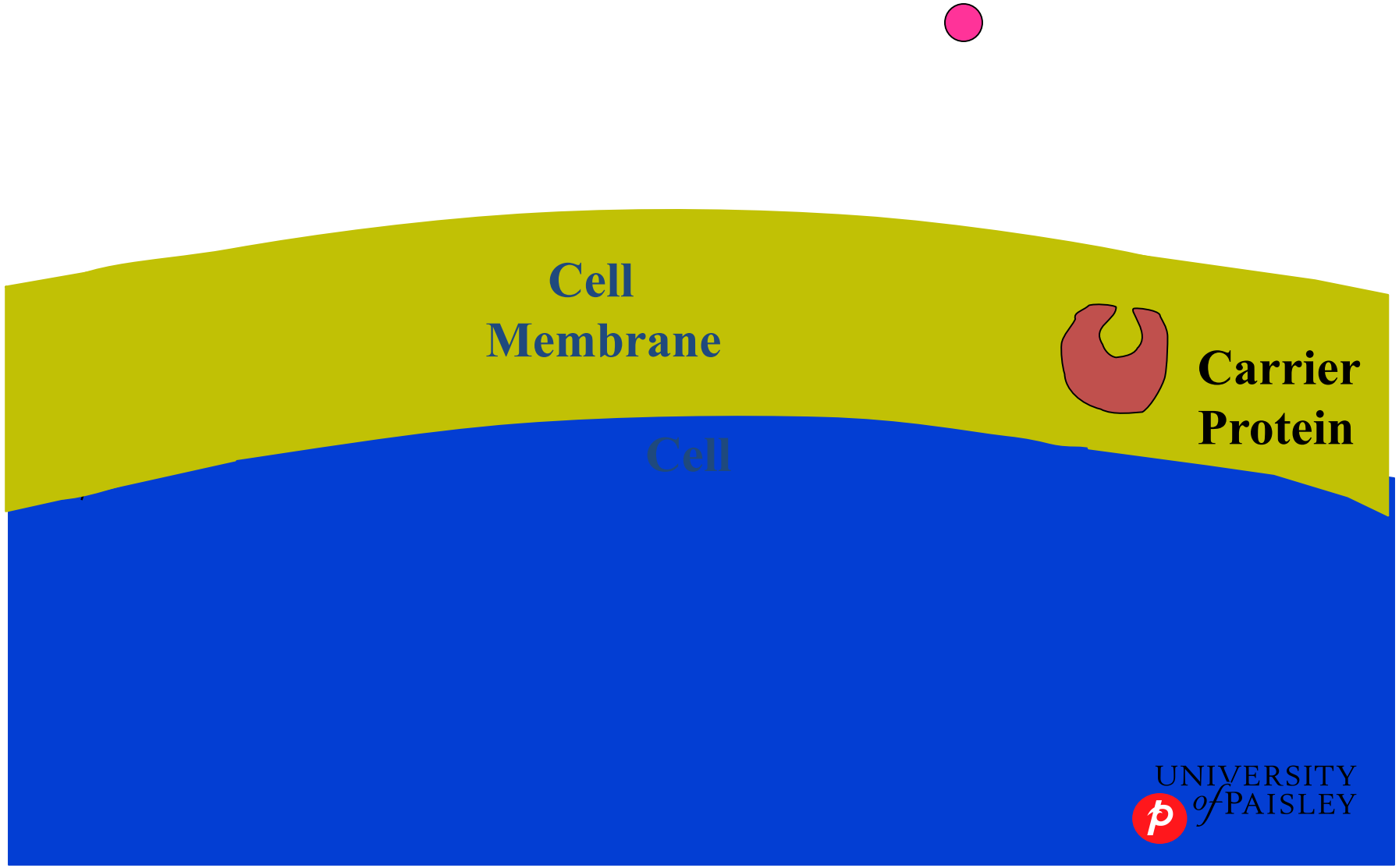
Dopamine

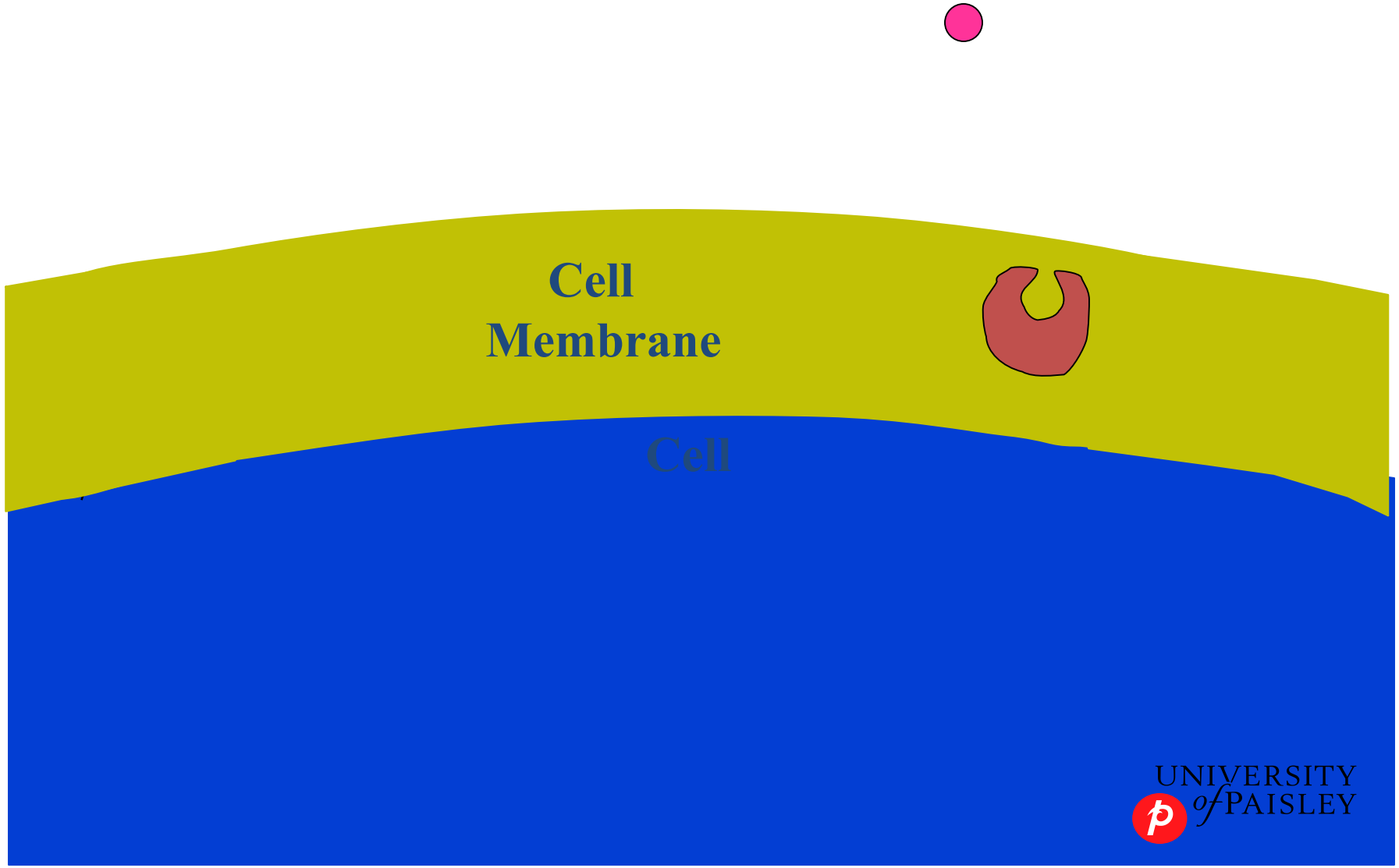
- **Useful in treating Parkinson's Disease**
- **Too polar to cross cell membranes and BBB**

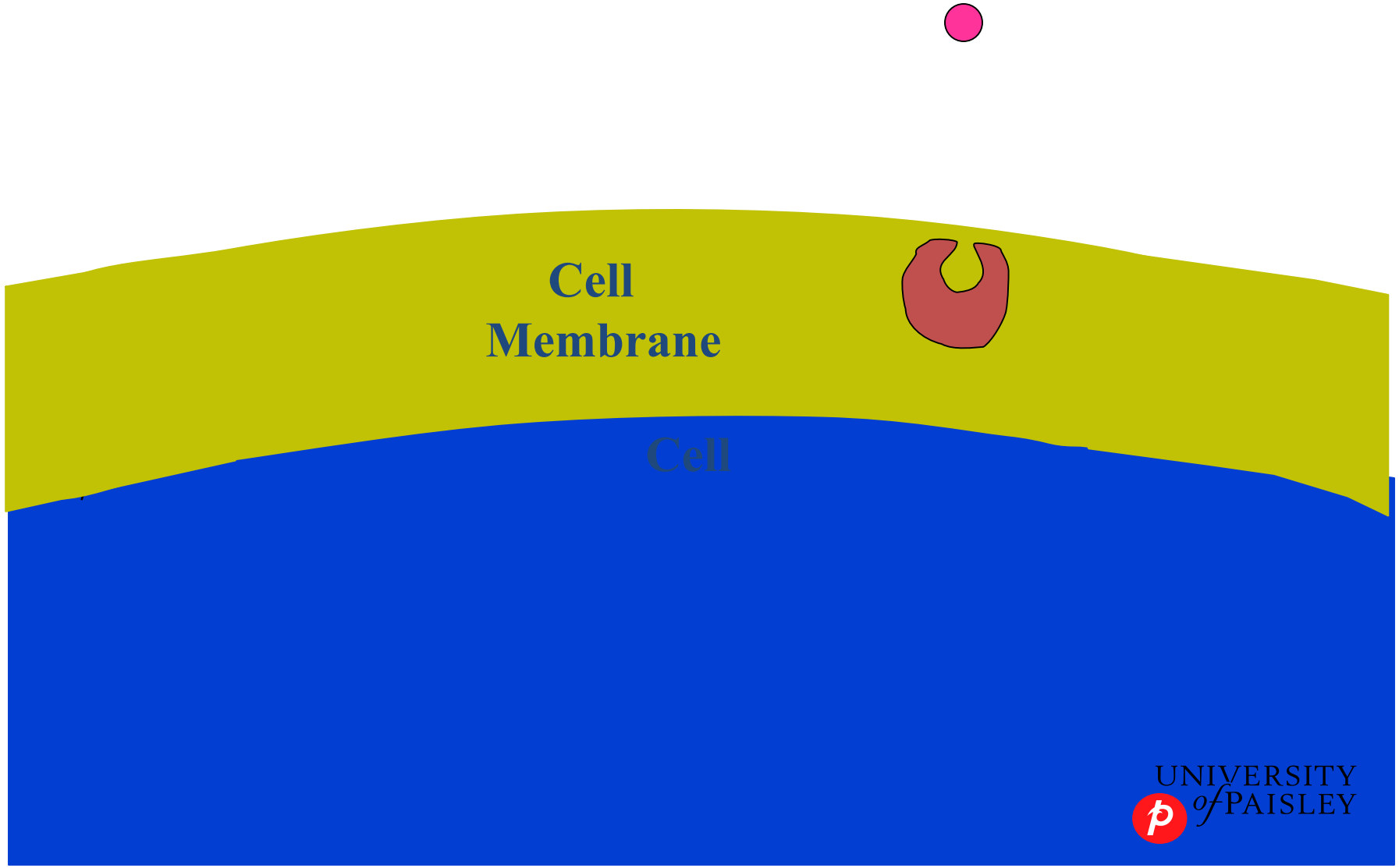


Levodopa

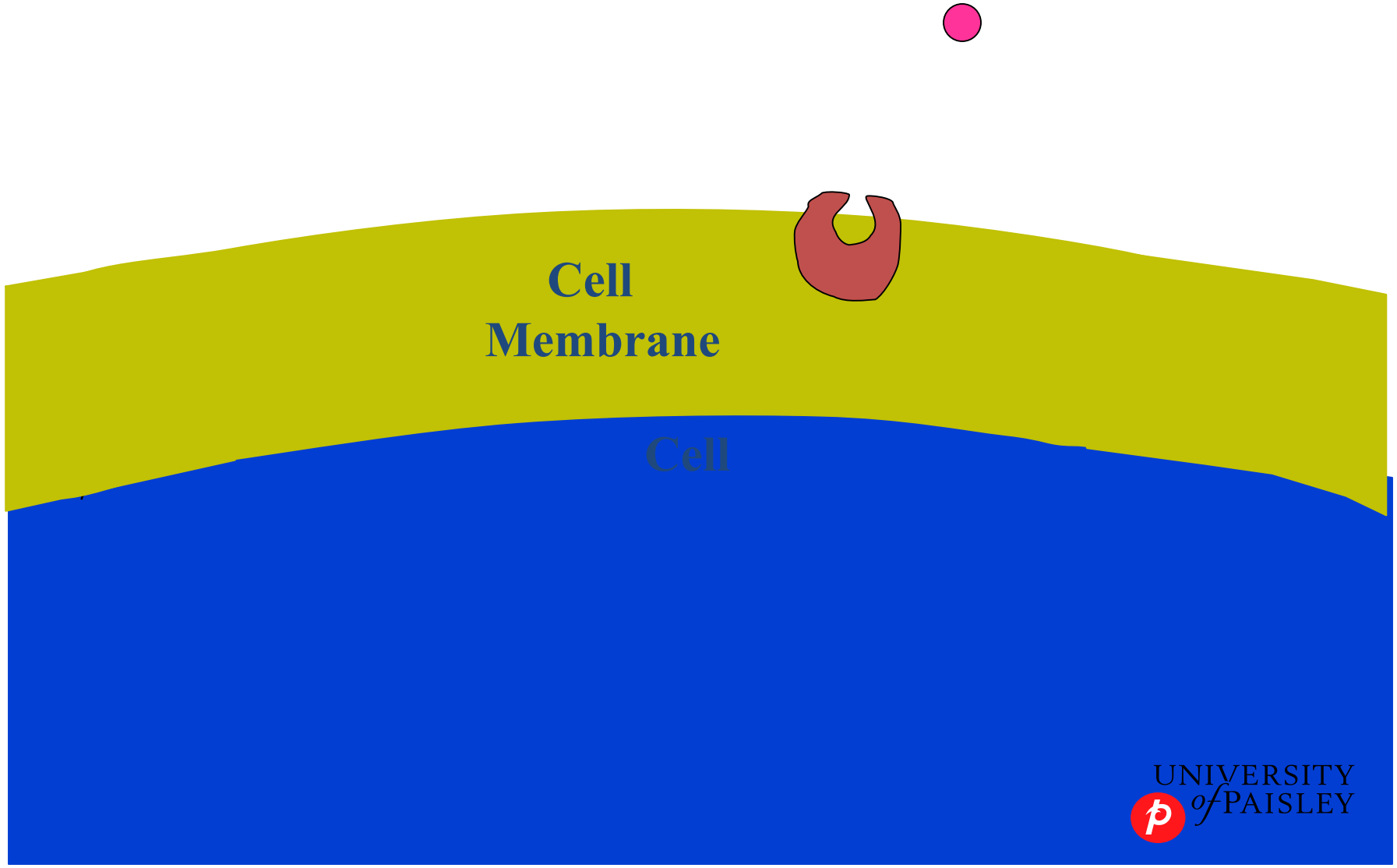
- **More polar but is an amino acid**
- **Carried across cell membranes by carrier proteins for amino acids**
- **Decarboxylated in cell to dopamine**


Cell
Membrane
Carrier
Protein
Cell
UNIVERSITY
of PAISLEY

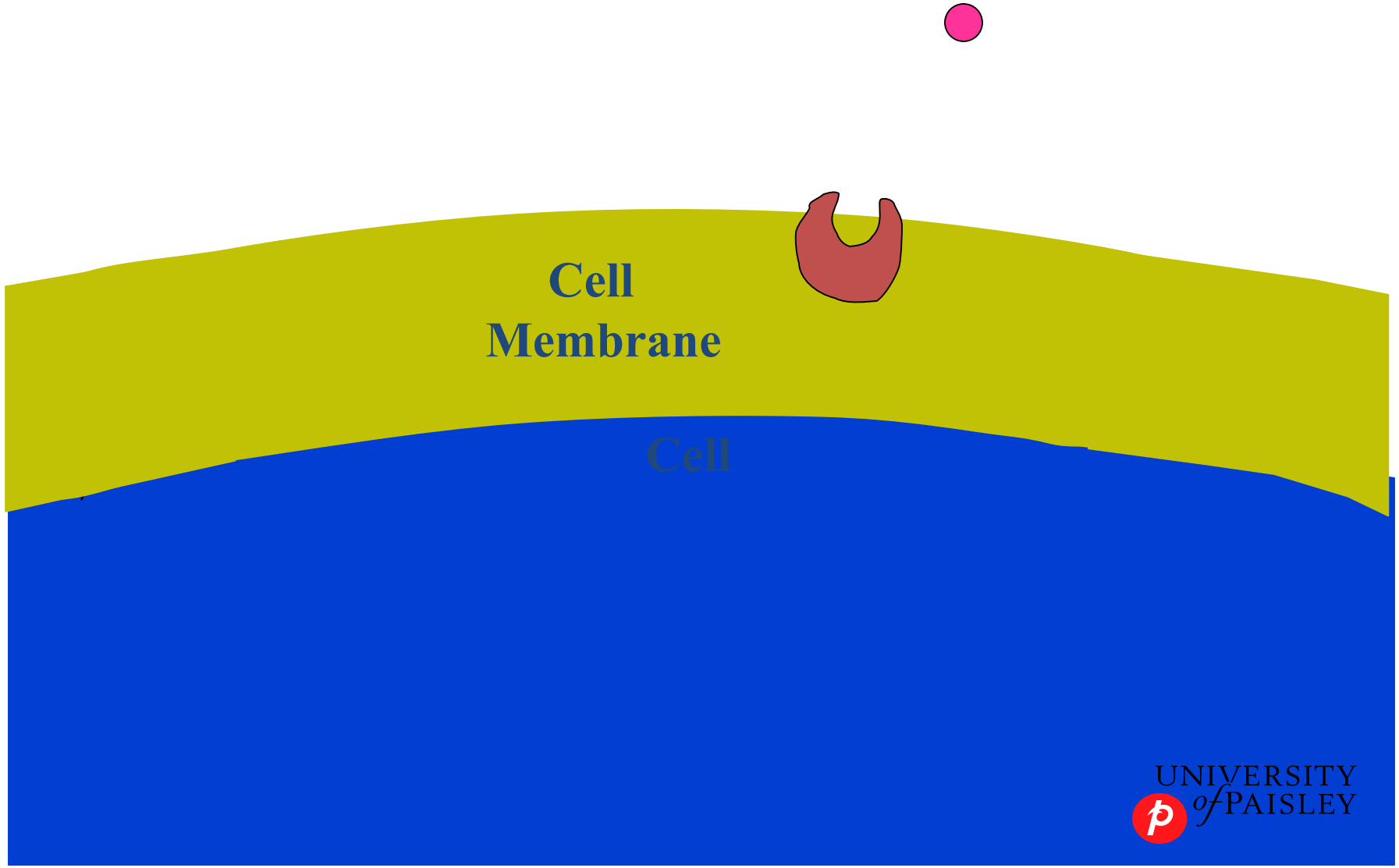

Cell
Membrane
Cell
UNIVERSITY
of PAISLEY

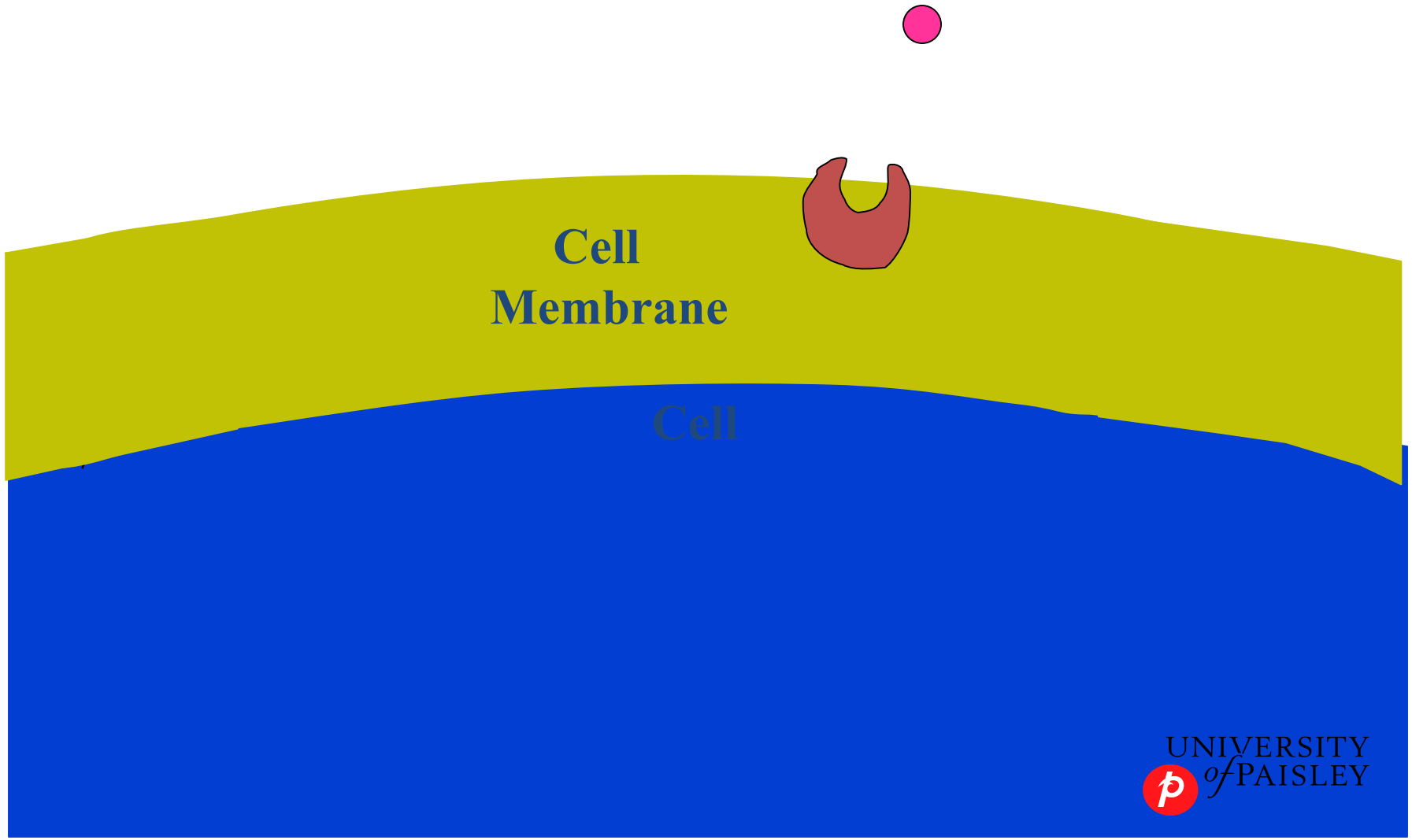


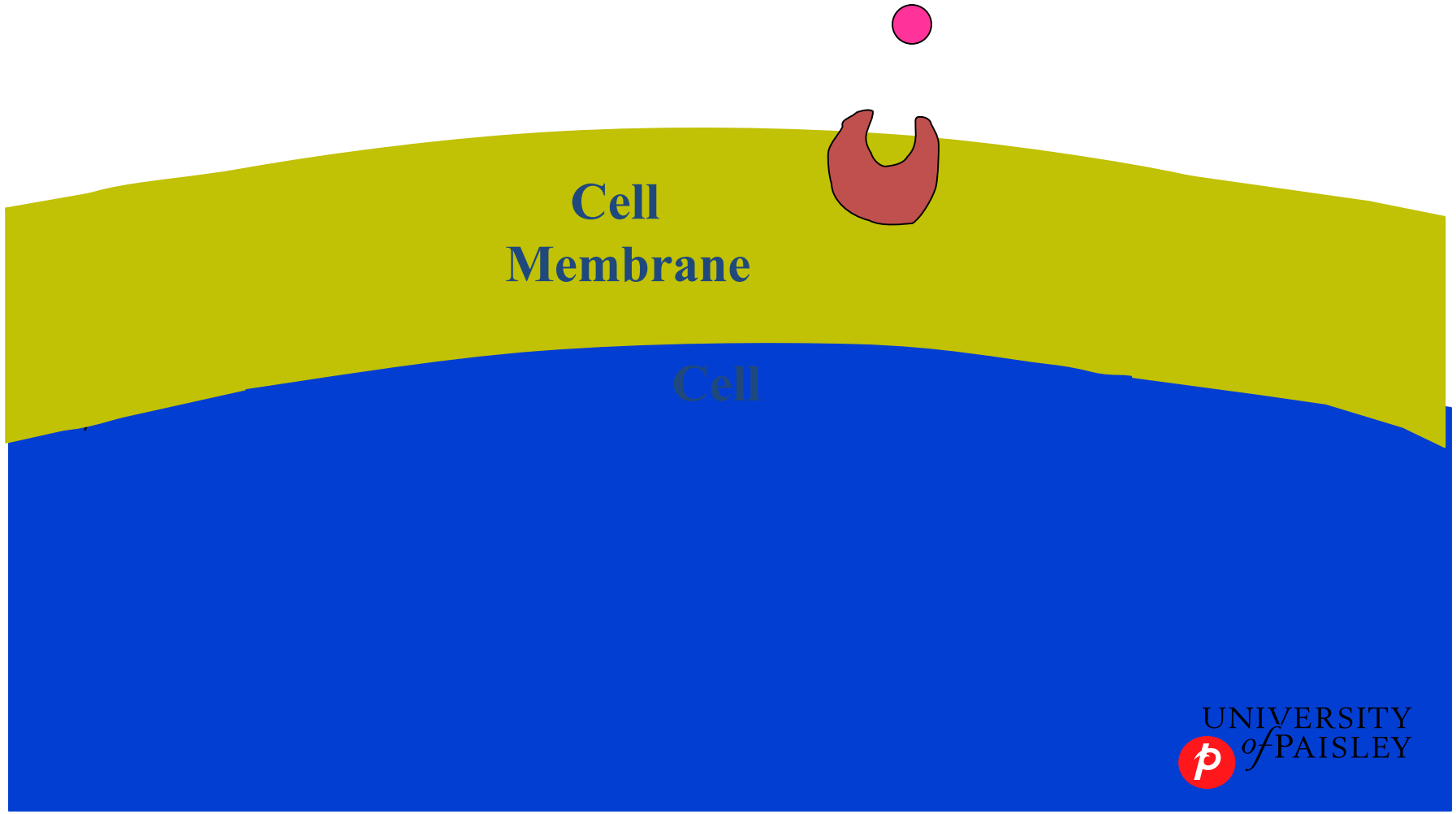
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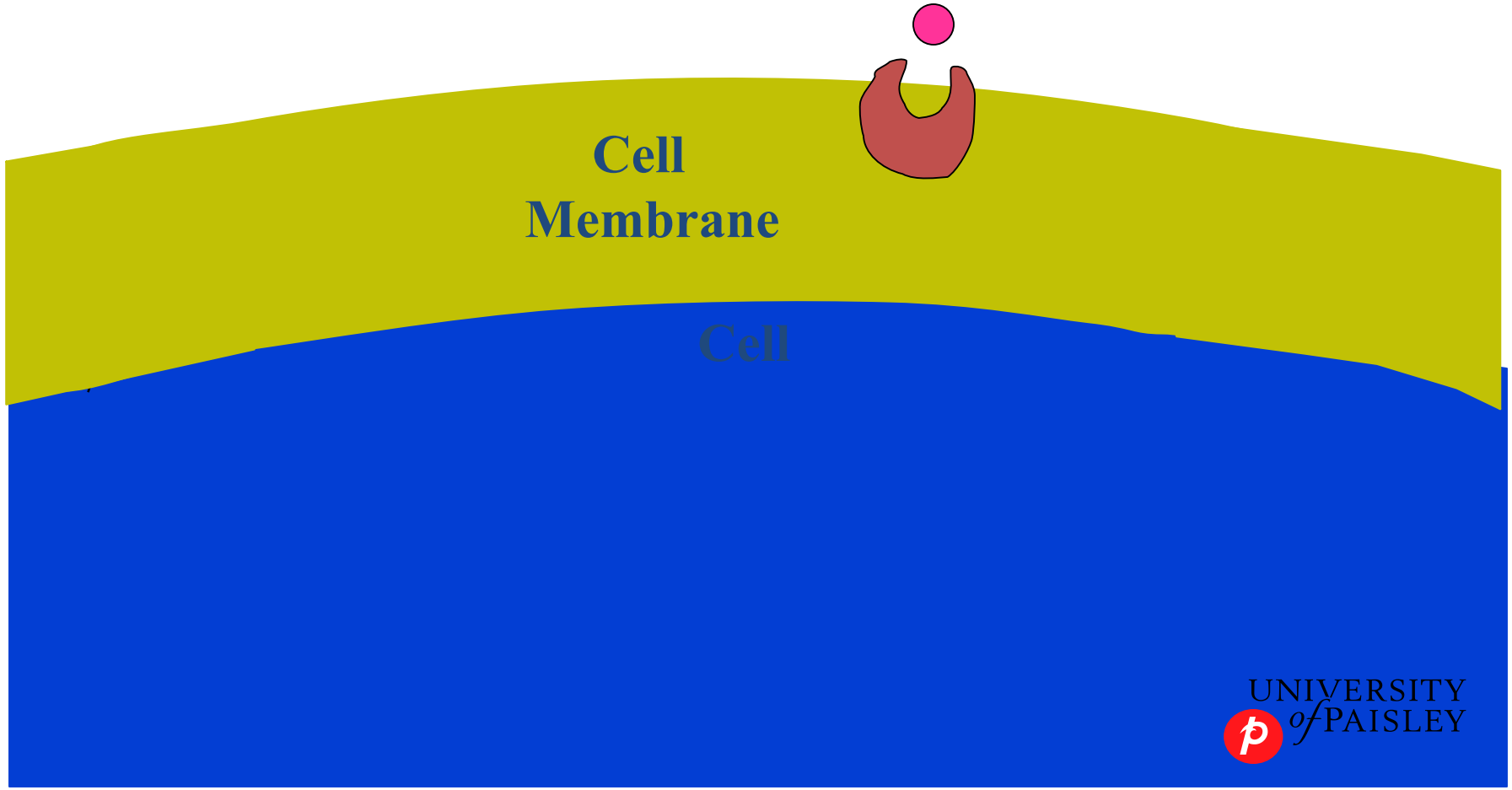
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Cell
Membrane
Cell

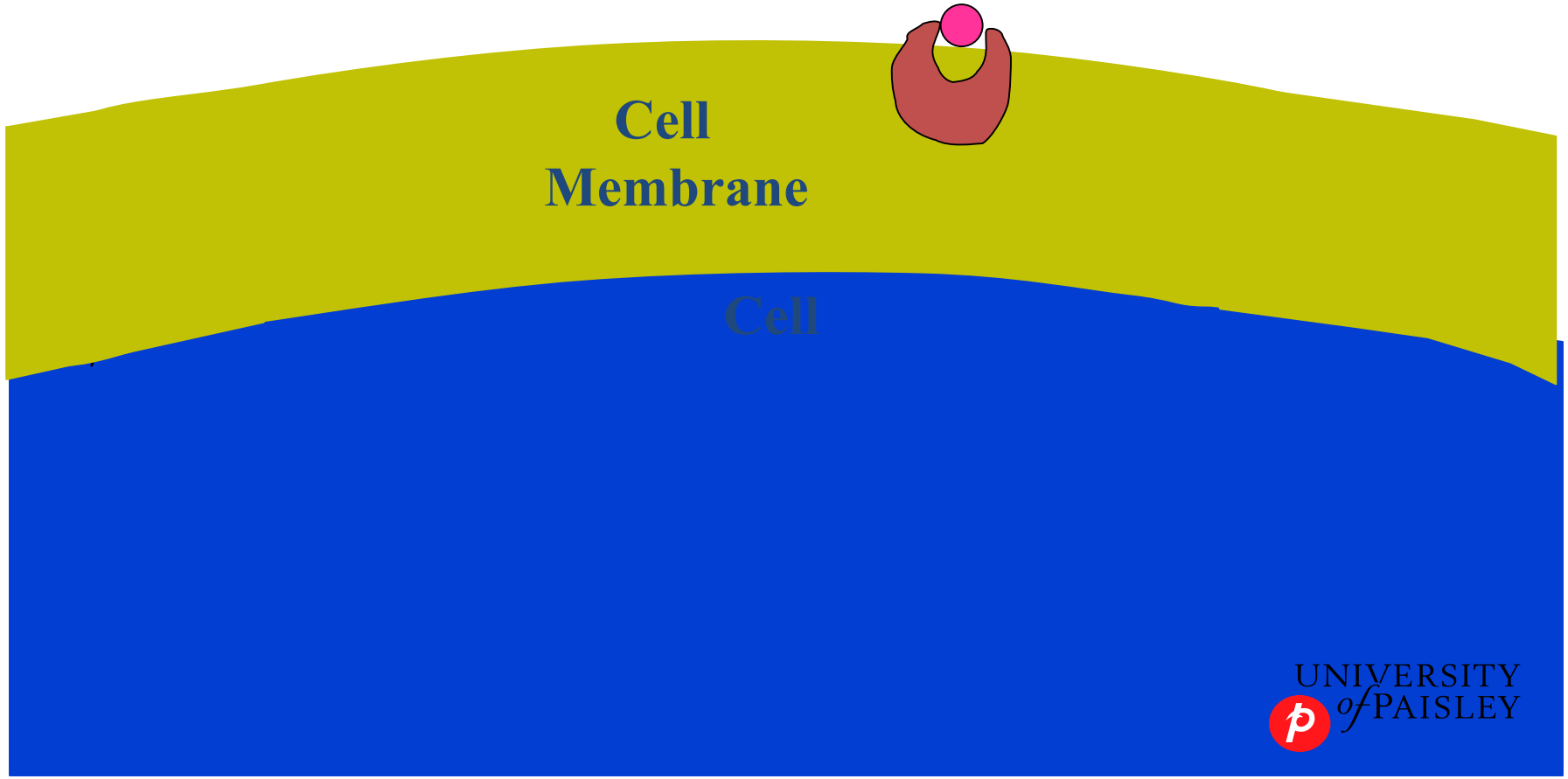

Cell
Membrane
Cell



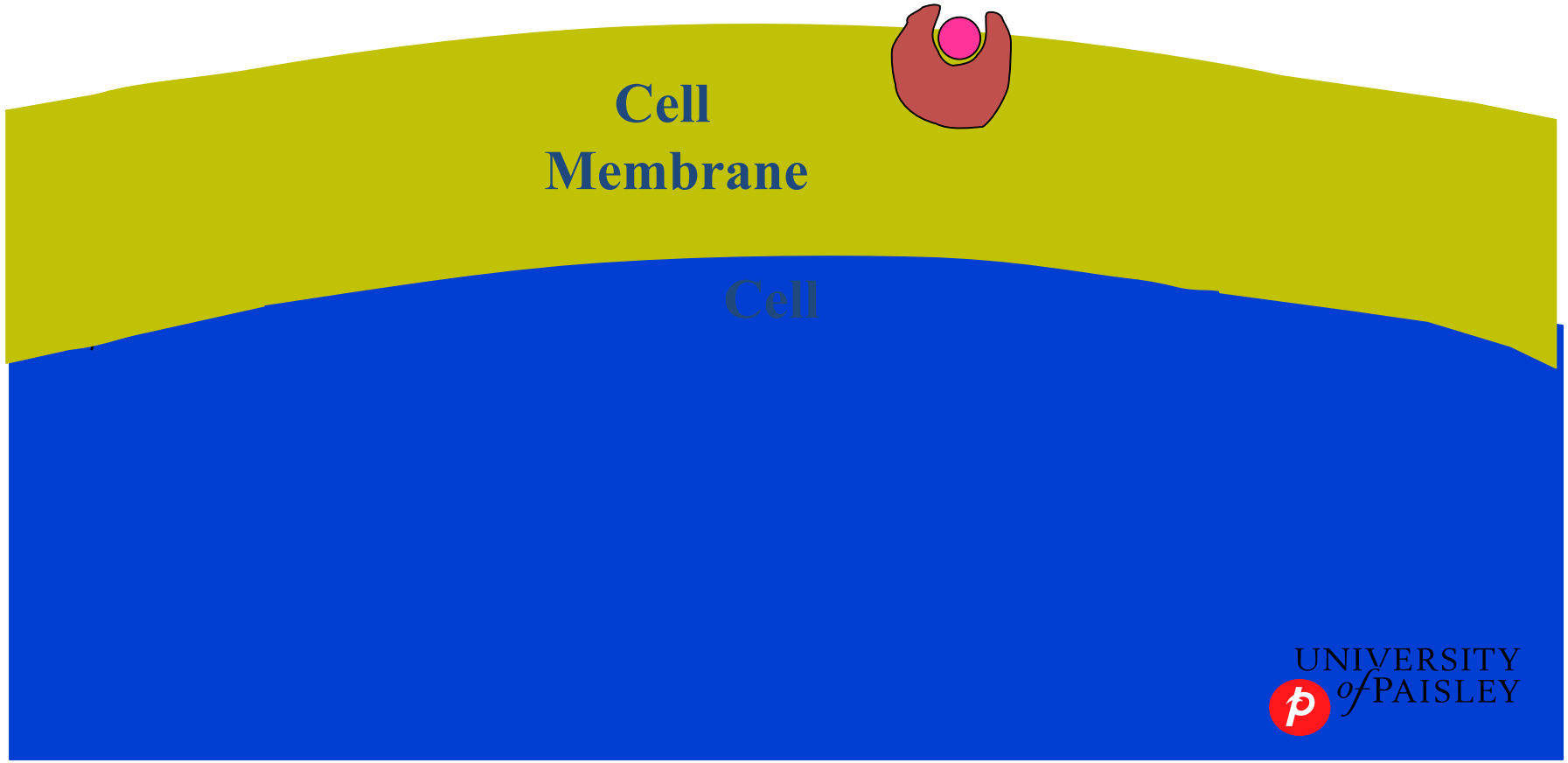
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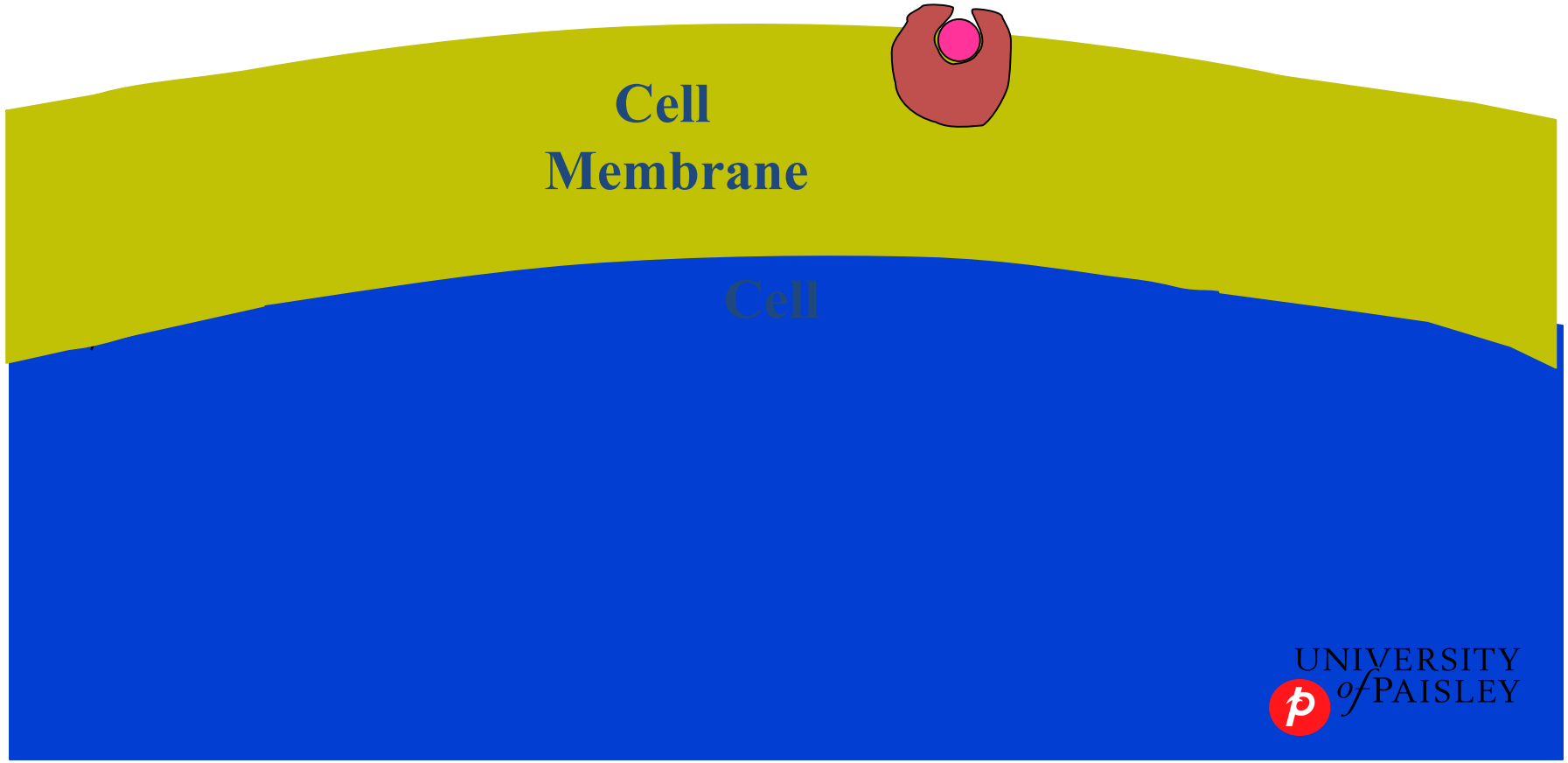
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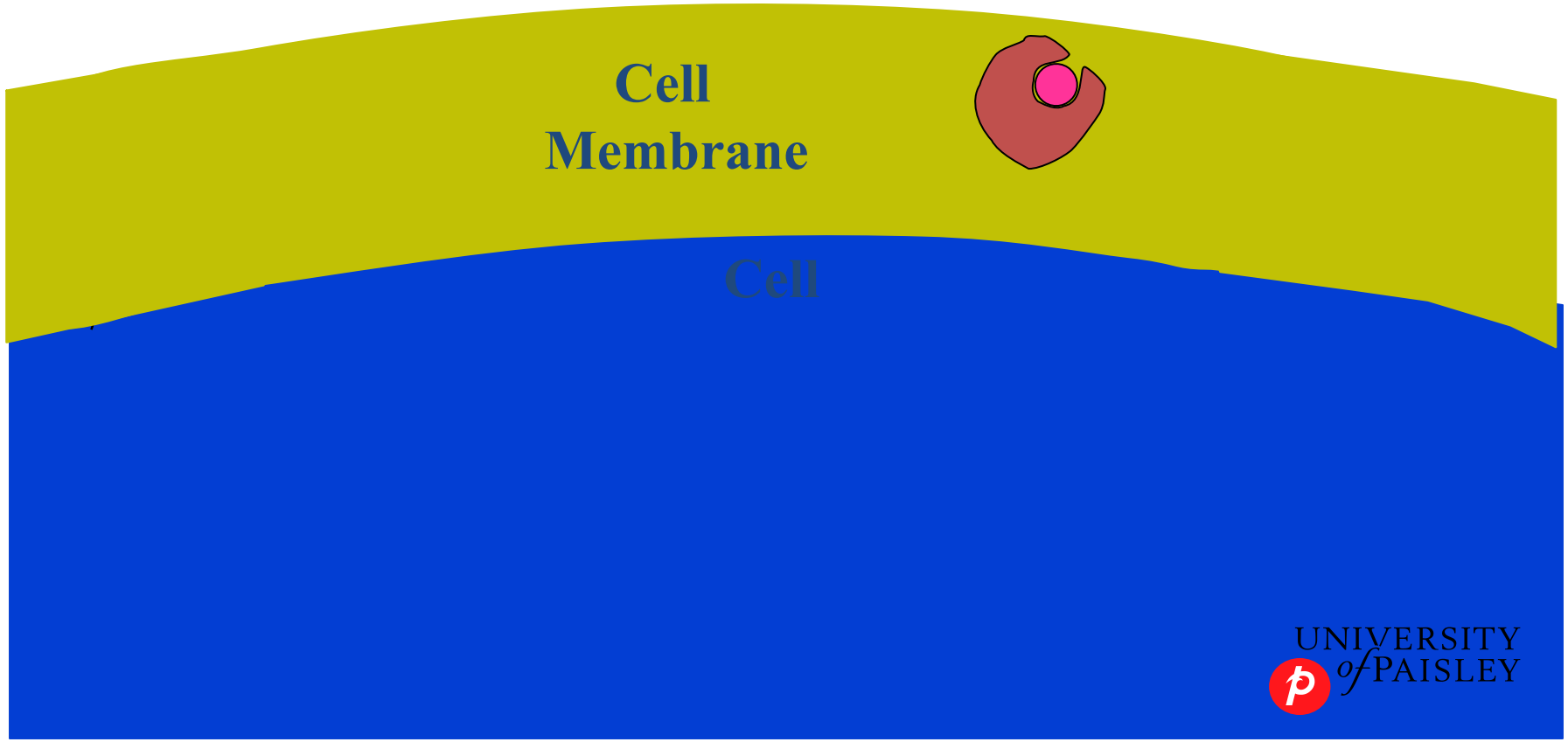


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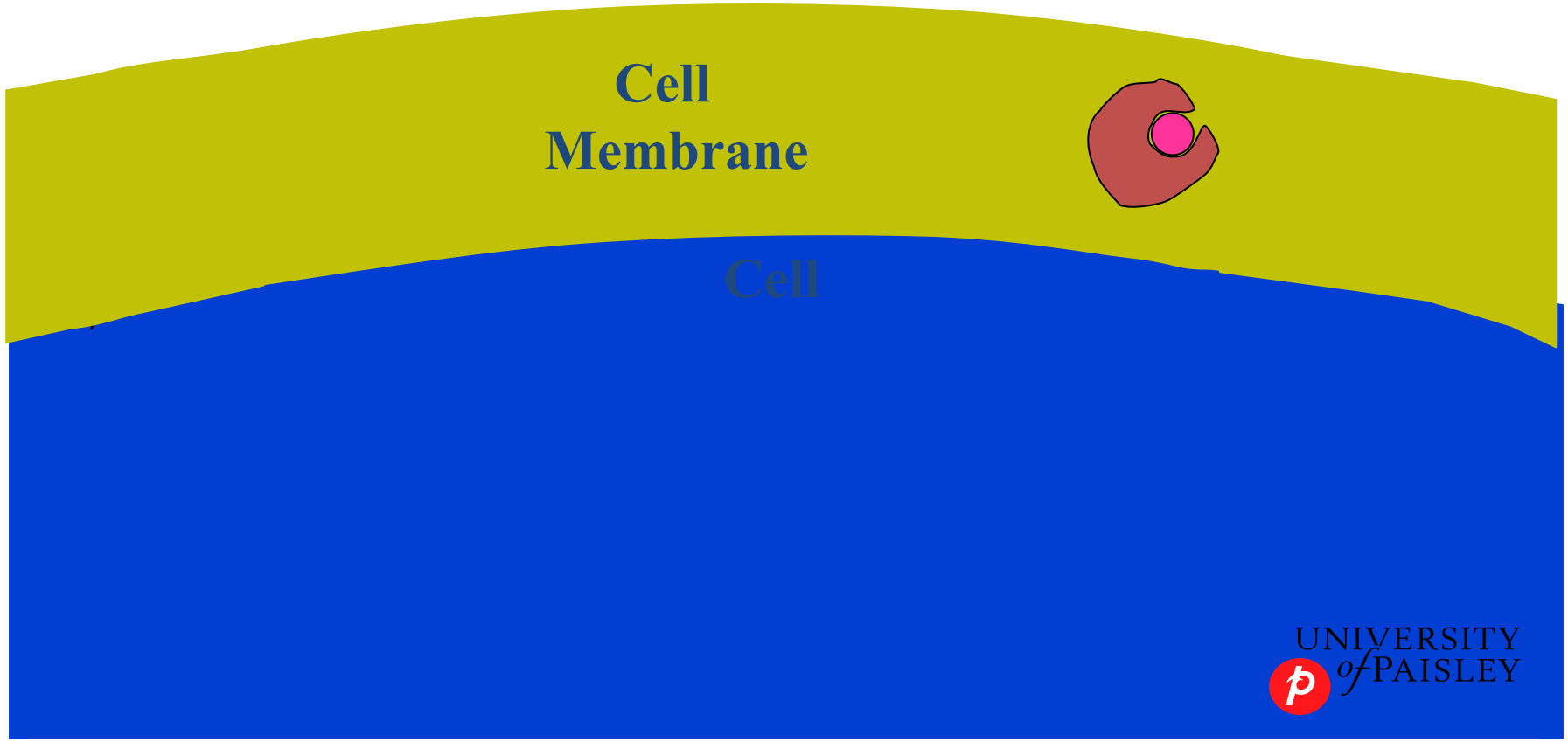


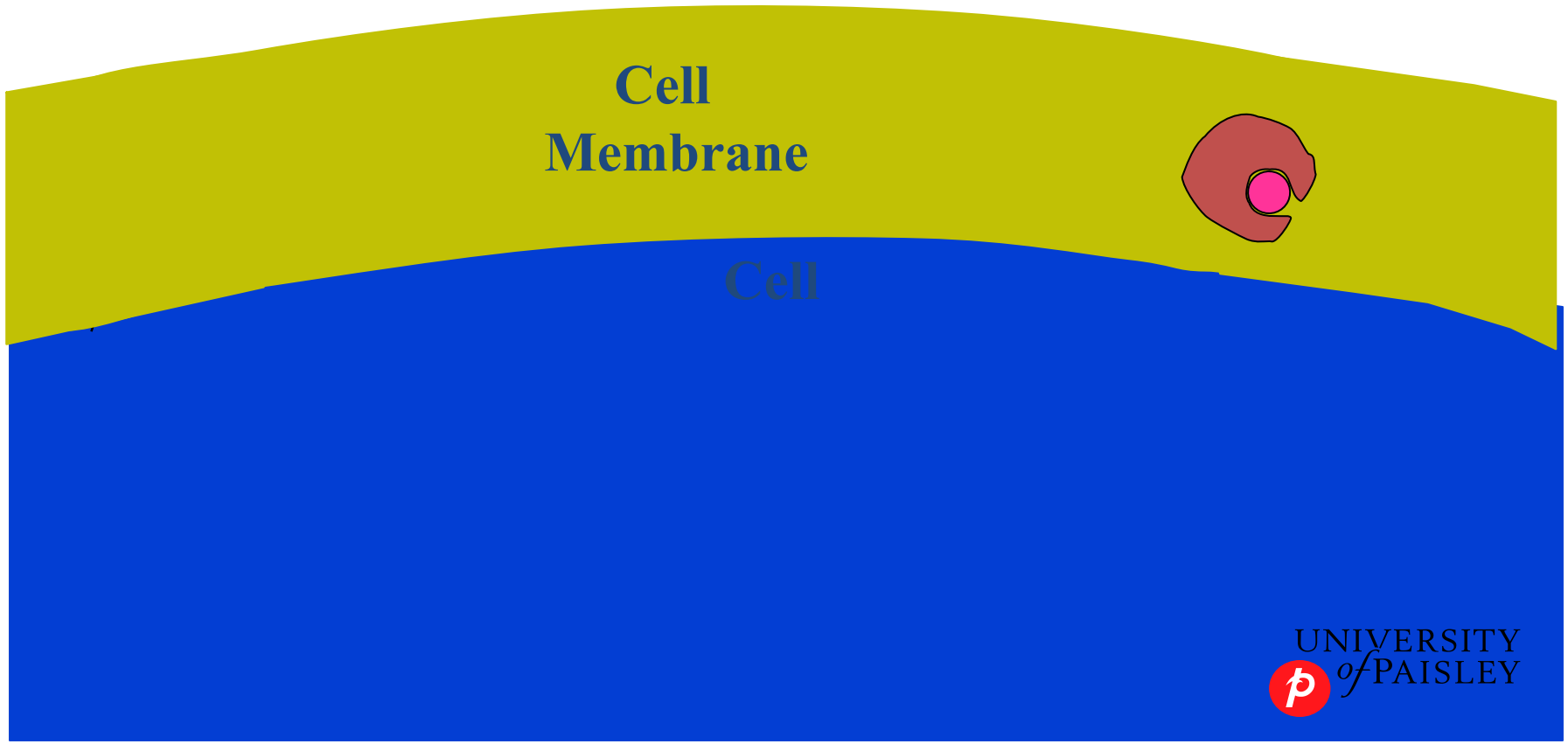
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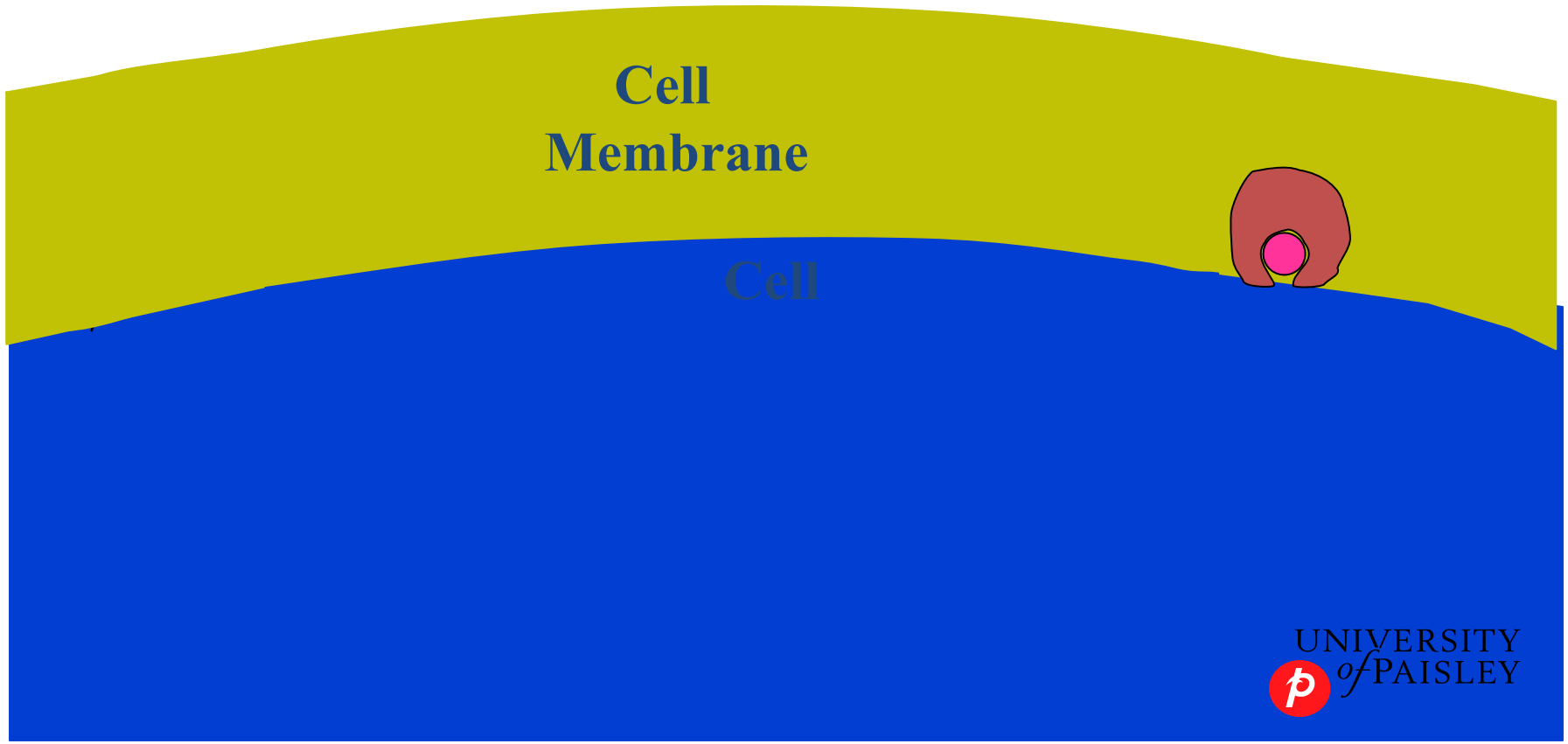

Cell
Membrane
Cell

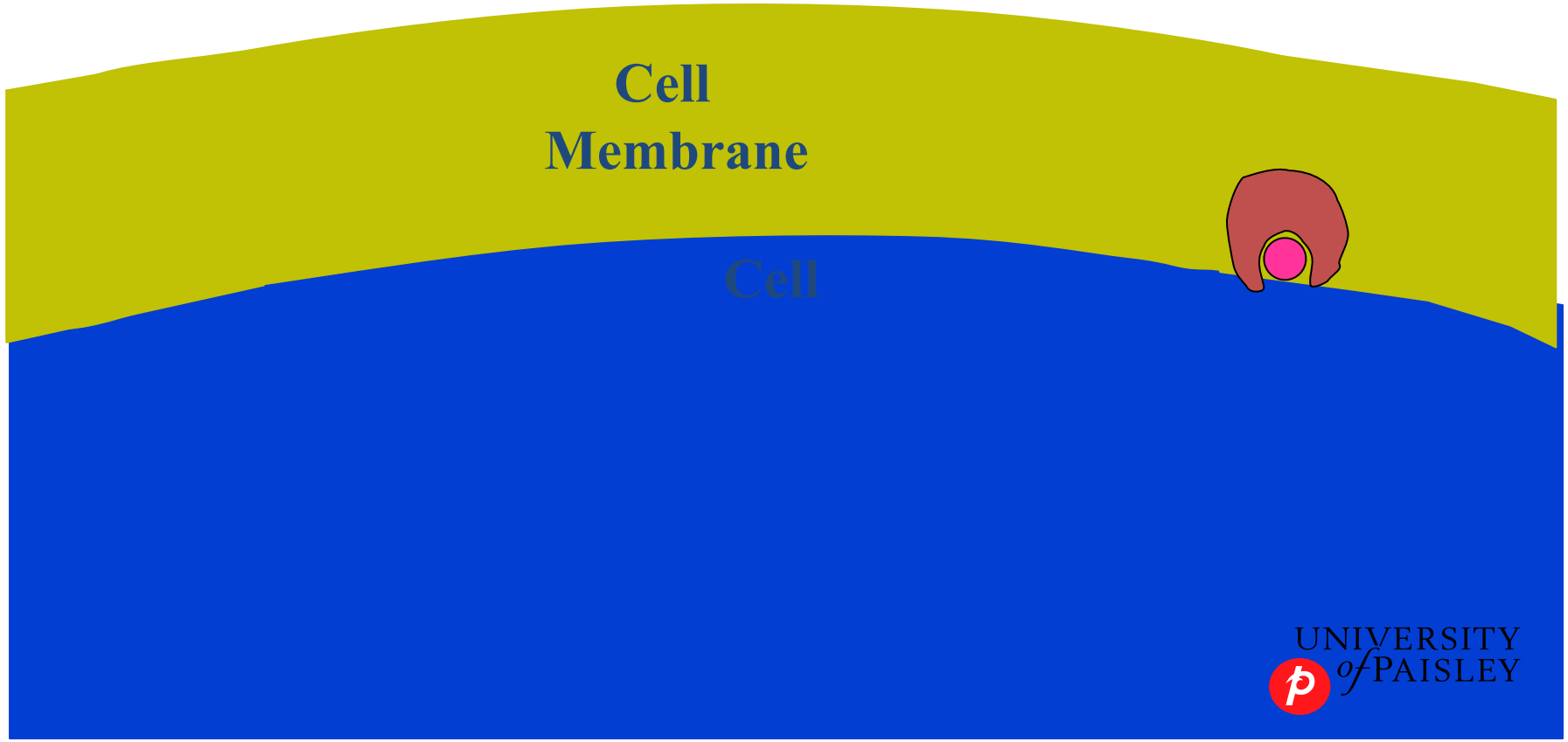


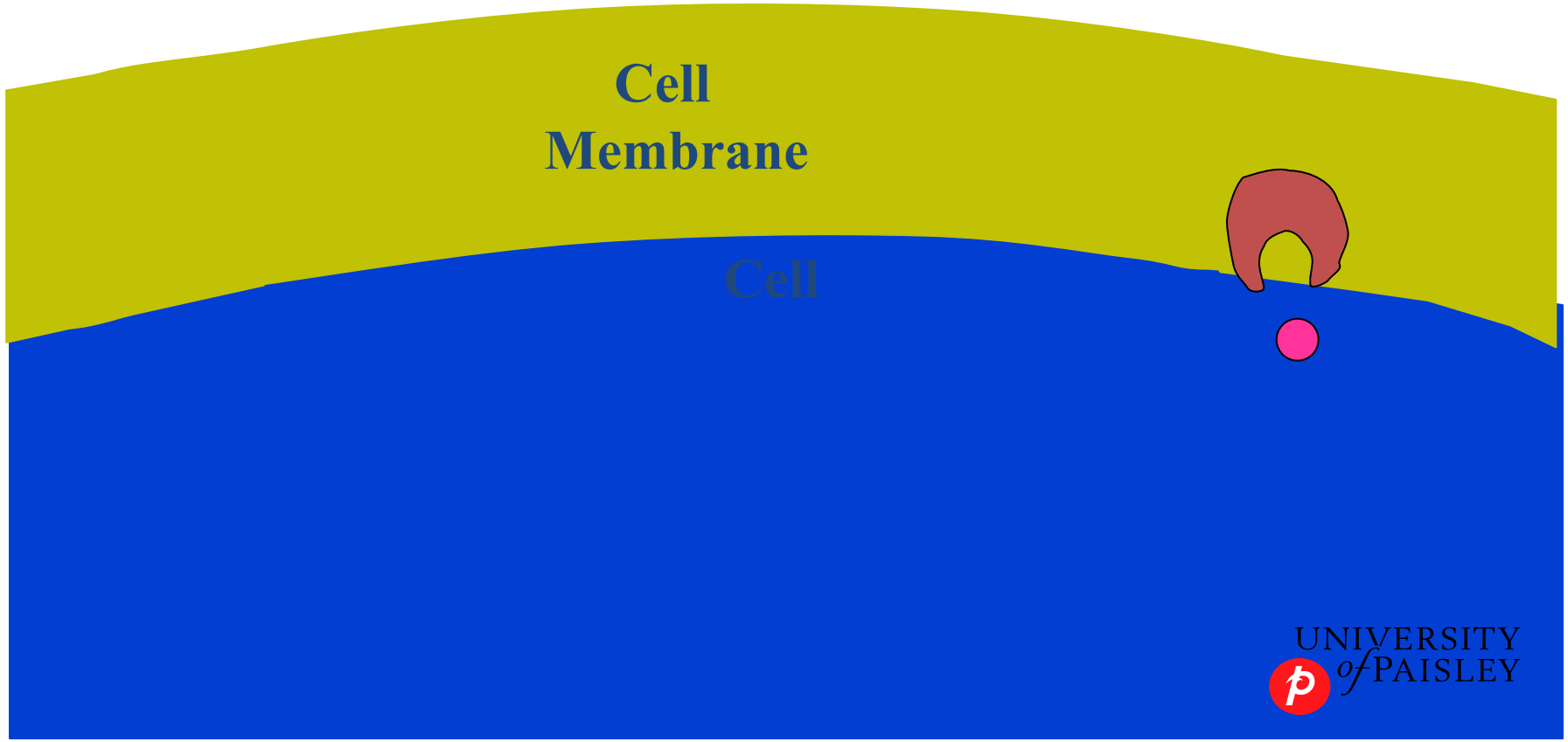
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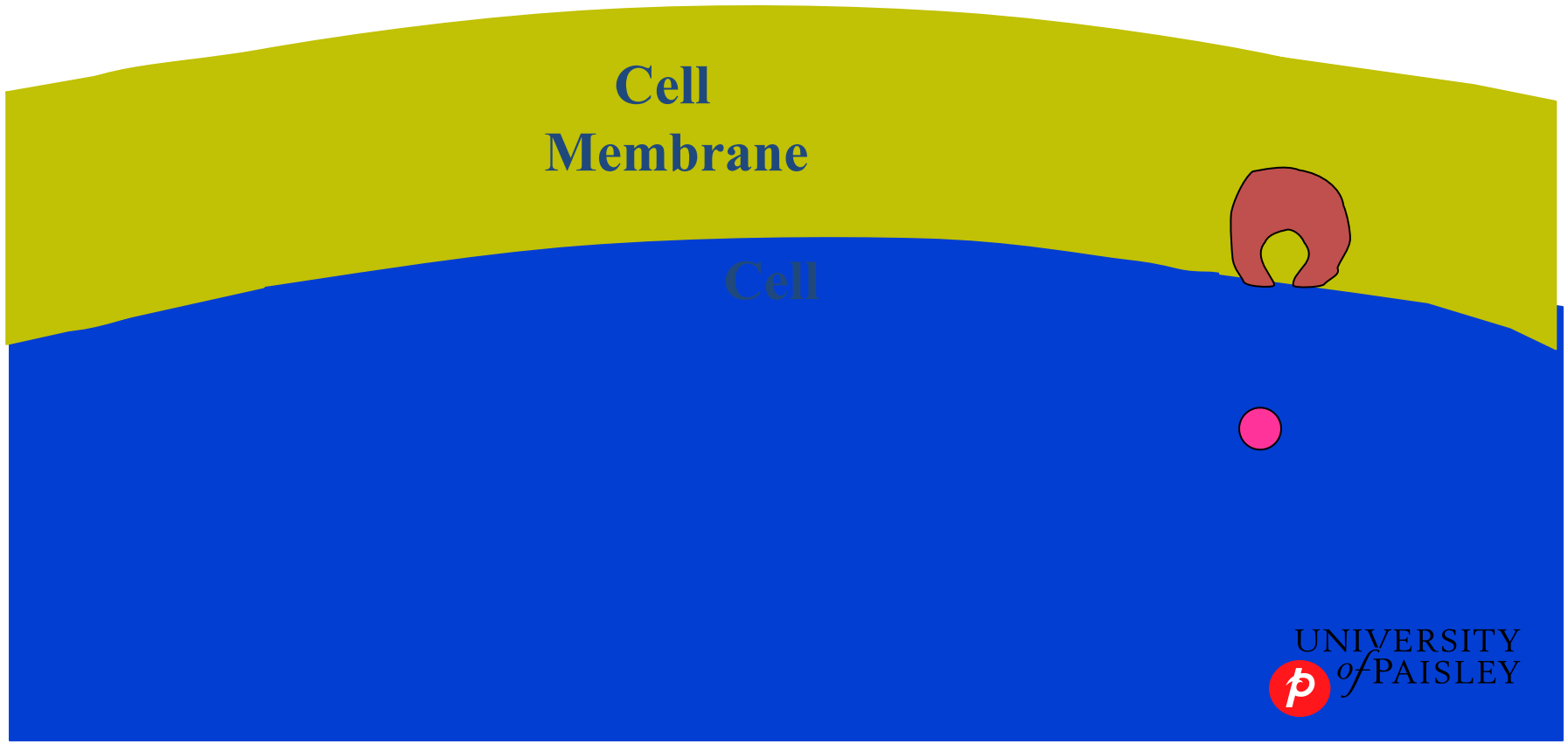

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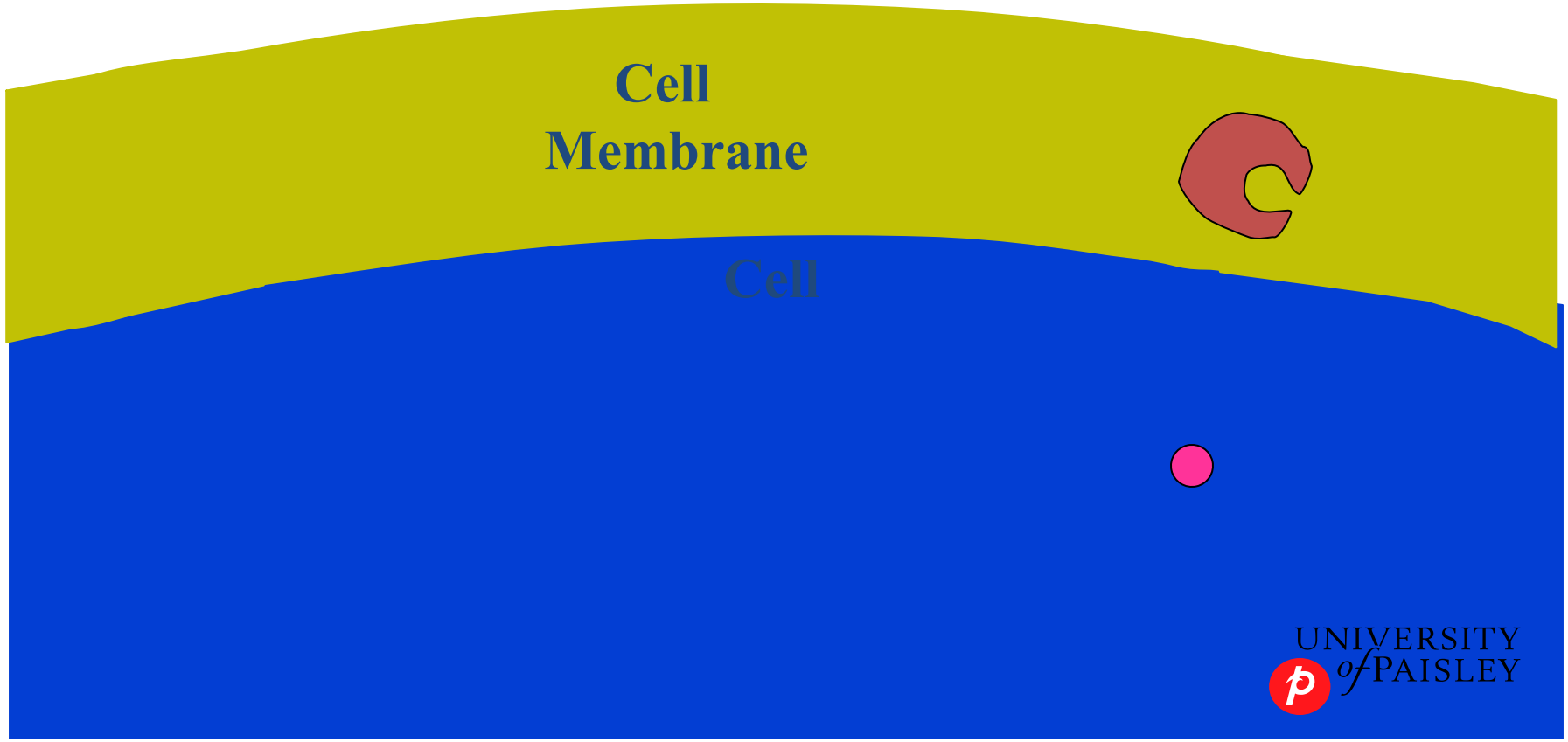

Cell
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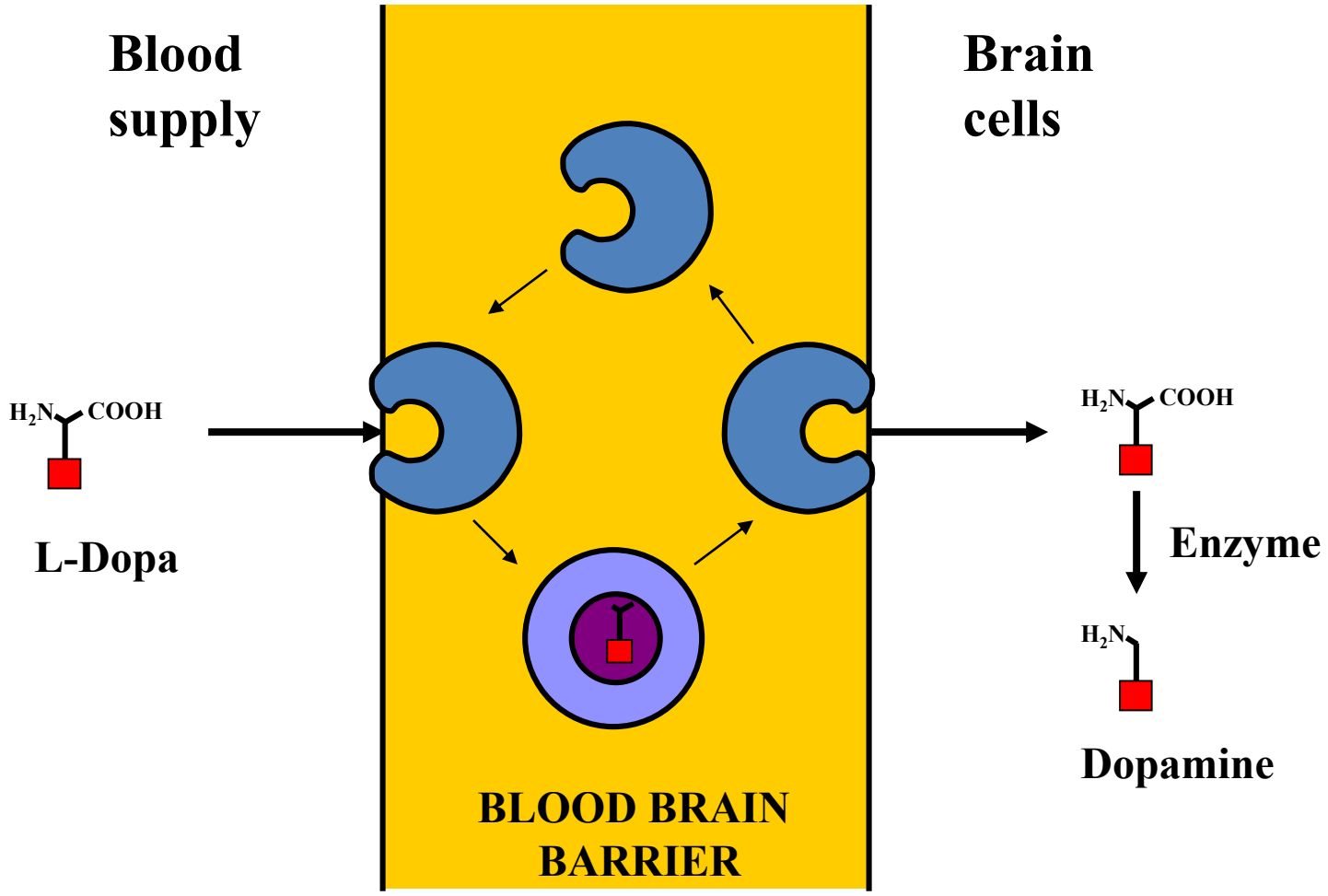

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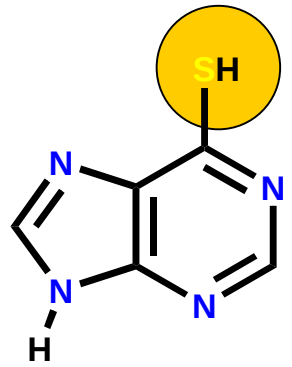

Cell
Membrane
Cell
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Cell
Membrane
Cell
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Blood supply
Brain cells
H_2N COOH
L-Dopa
BLOOD BRAIN BARRIER
H_2N COOH
Enzyme
H_2N
Dopamine

5.2 Prodrugs to prolong activity

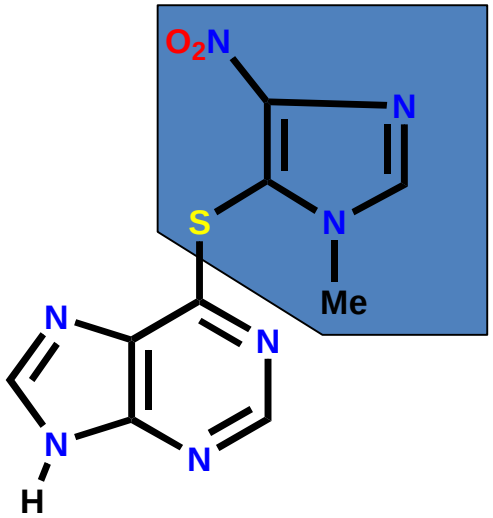
Mask polar groups



6-Mercaptopurine

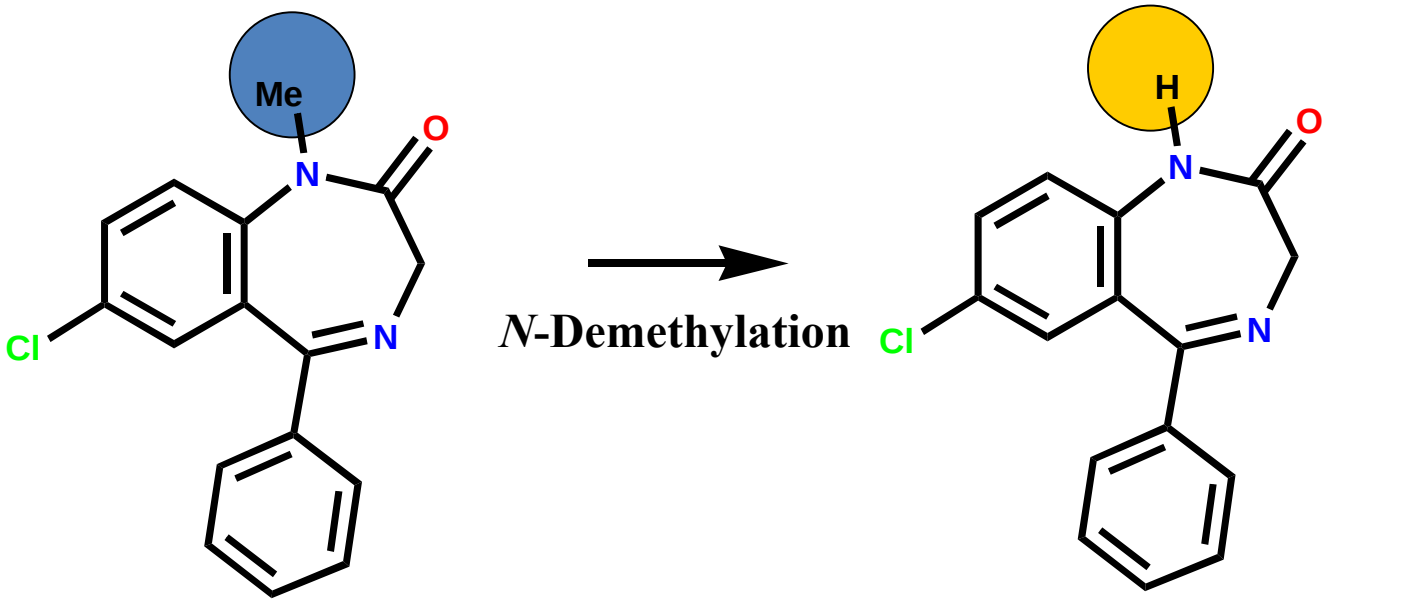
(suppresses immune response)

- **Short lifetime - eliminated too quickly**



Azathioprine

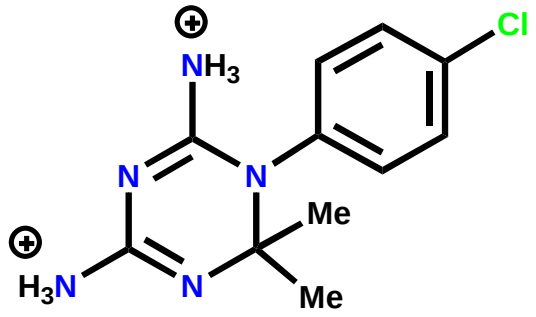
- **Slow conversion to 6-mercaptopurine**
- **Longer lifetime**



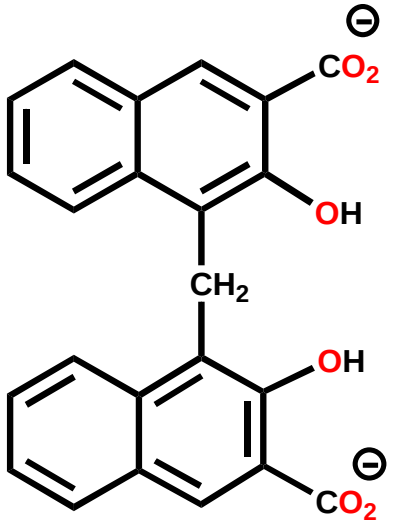
Valium prodrug **Nordazepam**

Add hydrophobic groups

- Drug concentrated in fat tissue
- Slow removal of hydrophobic group
- Slow release into blood supply

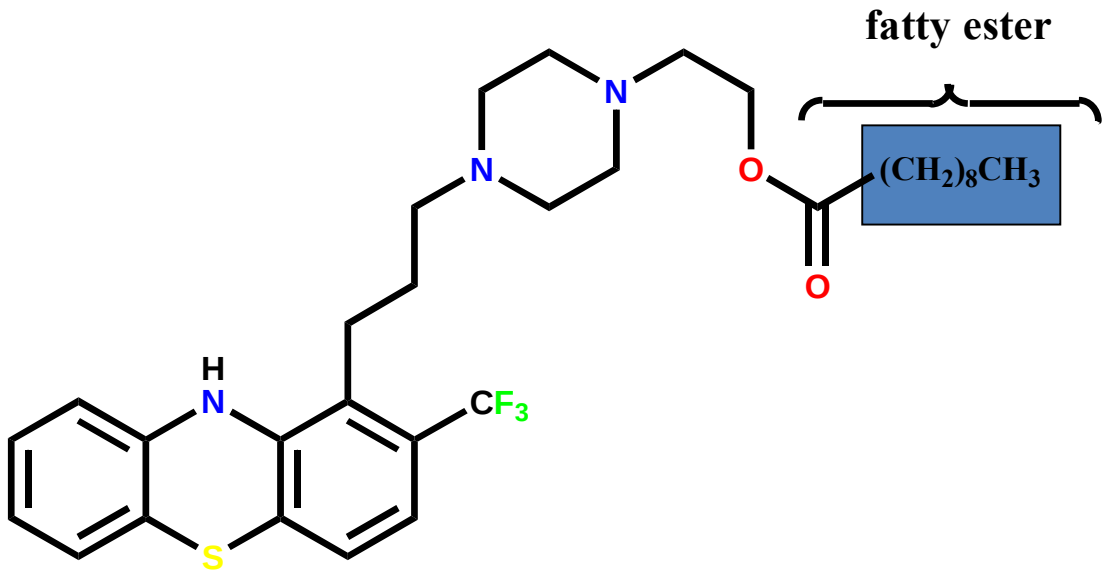


Cycloguanil



Pamoate

Lipophilic



Hydrophobic esters of fluphenazine (antipsychotic)

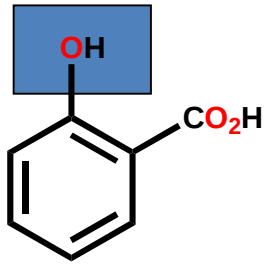
- Given by intramuscular injection
- Concentrated in fatty tissue
- Slowly released into the blood supply
- Rapidly hydrolysed in the blood supply

5.3 Prodrugs to mask toxicity and side effects

- Mask groups responsible for toxicity/side effects
- Used when groups are important for activity

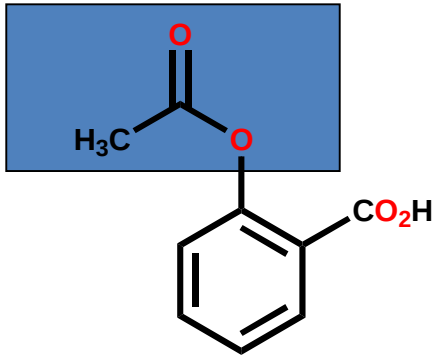
Example:

Aspirin for salicylic acid



Salicylic acid

- Analgesic, but causes stomach ulcers due to phenol group



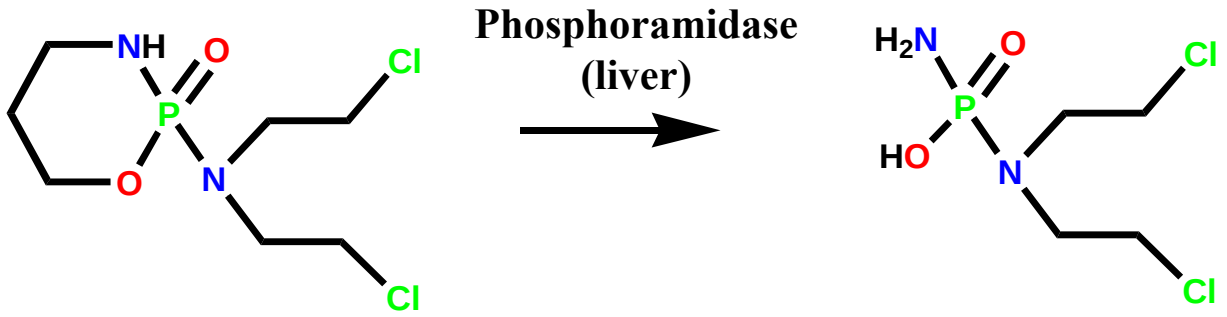
Aspirin

- Phenol masked by ester
- Hydrolysed in body

Example:

Cyclophosphoramide for phosphoramide mustard

(anticancer agent)



Cyclophosphoramide

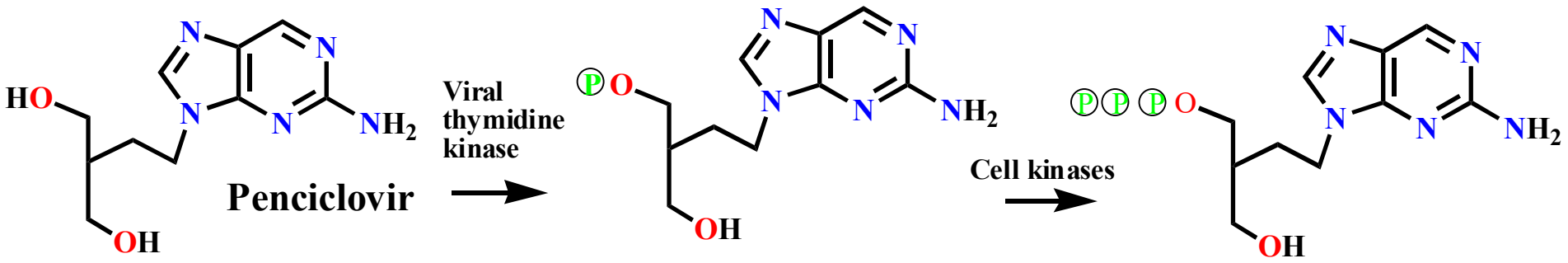
- Non toxic
- Orally active

Phosphoramide mustard

- Alkylating agent

Example:

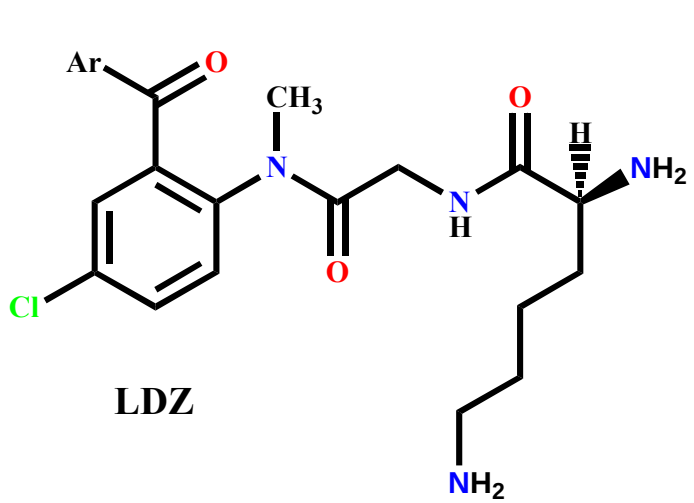
Antiviral drugs



activated by a viral enzyme which is only present in infected cells

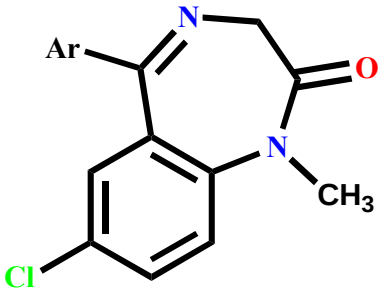
Example:

LDZ for diazepam



LDZ

a) Aminopeptidase
b) Cyclisation



Diazepam

\+ non-toxic lysine

LDZ

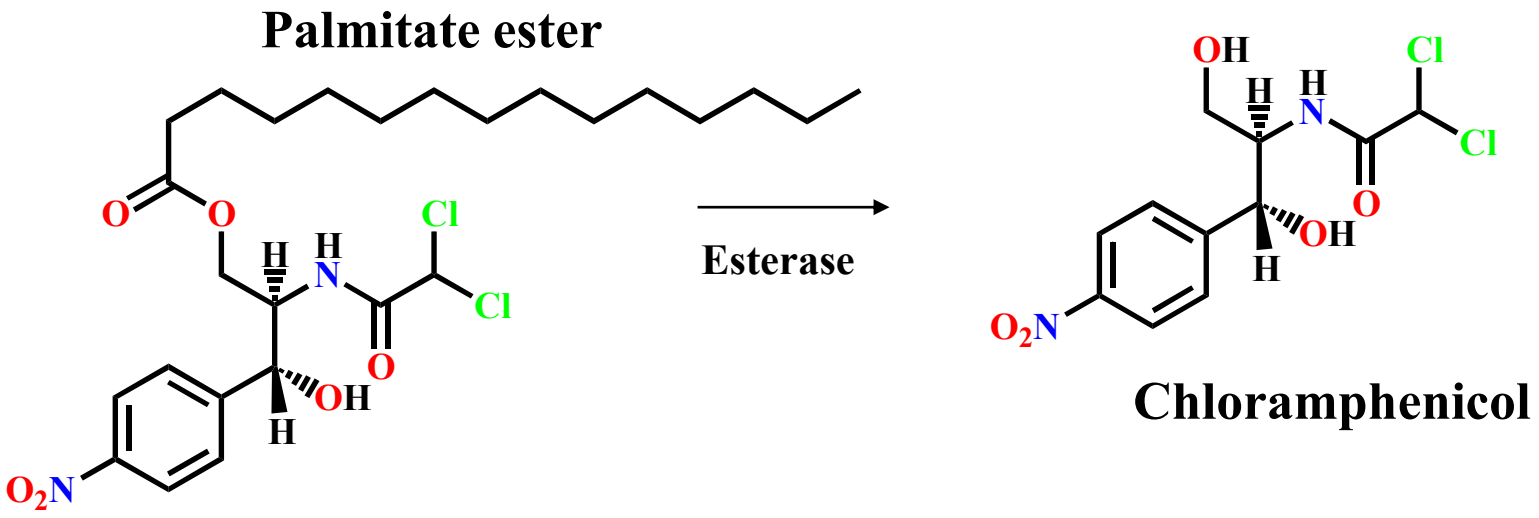
- **Avoids drowsy side effects of diazepam**

5.4 Prodrugs to lower water solubility

- Used to reduce solubility of foul tasting orally active drugs
- Less soluble on tongue
- Less revolting taste

Example:

Palmitate ester of chloramphenicol (antibiotic)

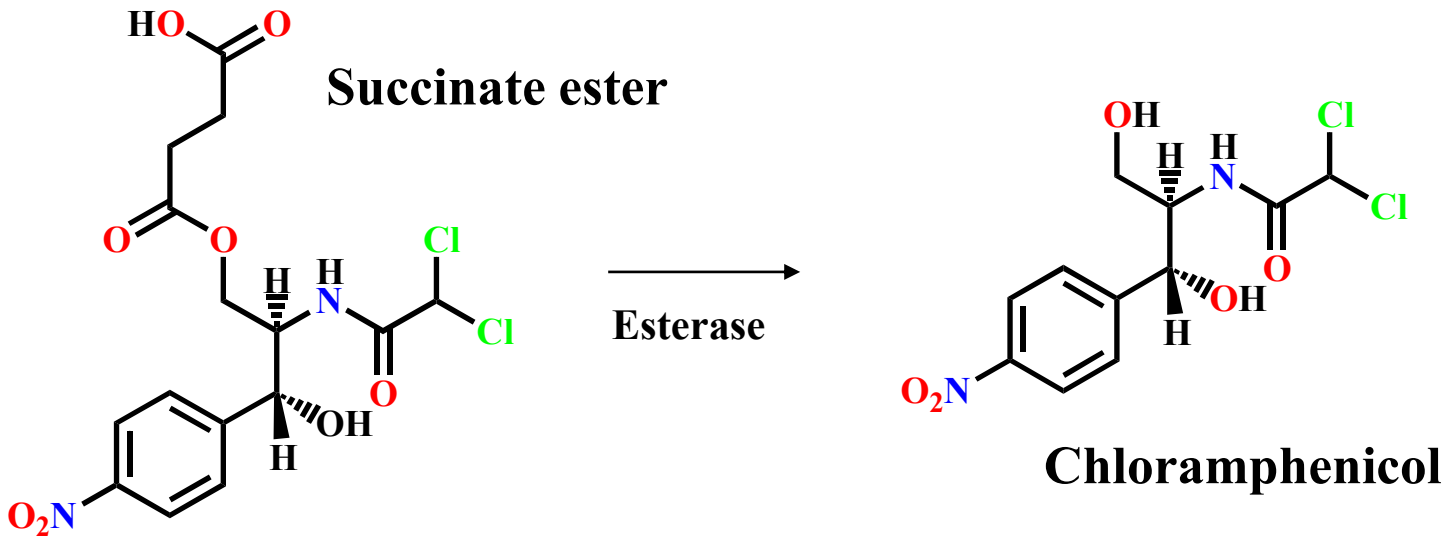


5.5 Prodrugs to increase water solubility

- Often used for i.v. drugs
- Allows higher concentration and smaller dose volume
- May decrease pain at site of injection

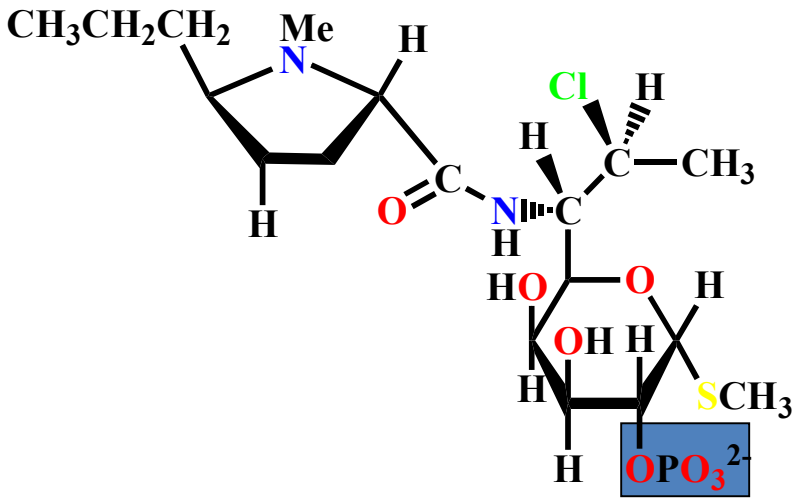
Example:

Succinate ester of chloramphenicol (antibiotic)



Example:

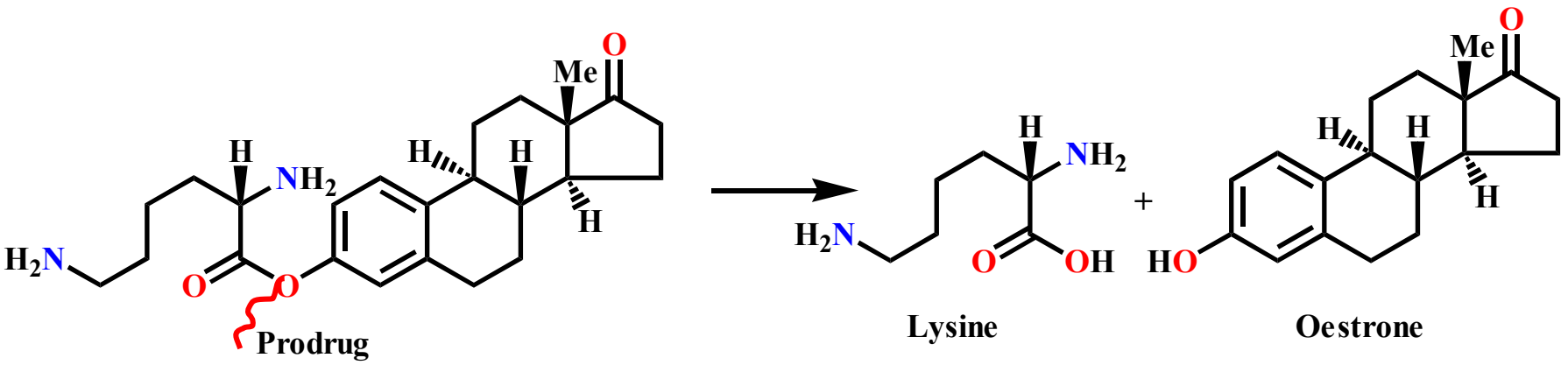
Phosphate ester of clindamycin (antibacterial)



- **Less painful on injection**

Example:

Lysine ester of oestrone

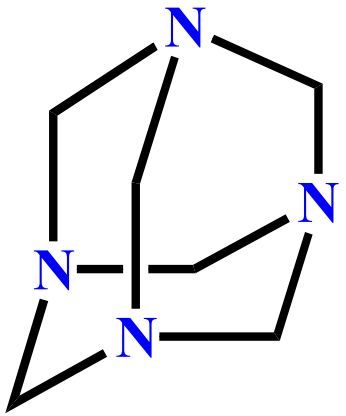


- Lysine ester of oestrone is better absorbed orally than oestrone
- Increased water solubility prevents formation of fat globules in gut
- Better interaction with the gut wall
- Hydrolysis in blood releases oestrone and a non toxic amino acid

5.6 Prodrugs used to target drugs

Example:

Hexamine

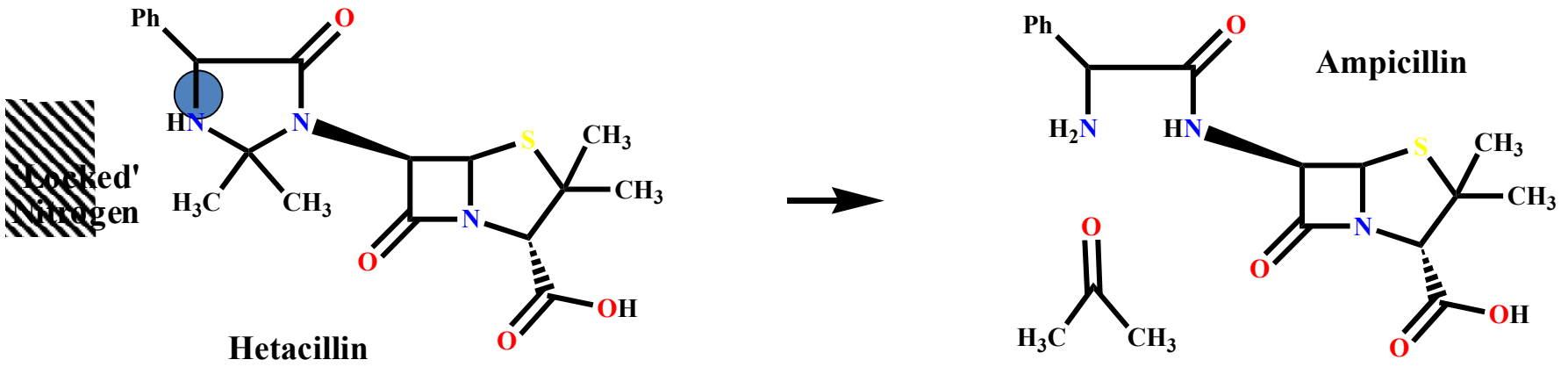


- Stable and inactive at pH >5
- Stable at blood pH
- Used for urinary infections where pH<5
- Degrades at pH<5 to form formaldehyde (antibacterial agent)

5.7 Prodrugs to increase chemical stability

Example:

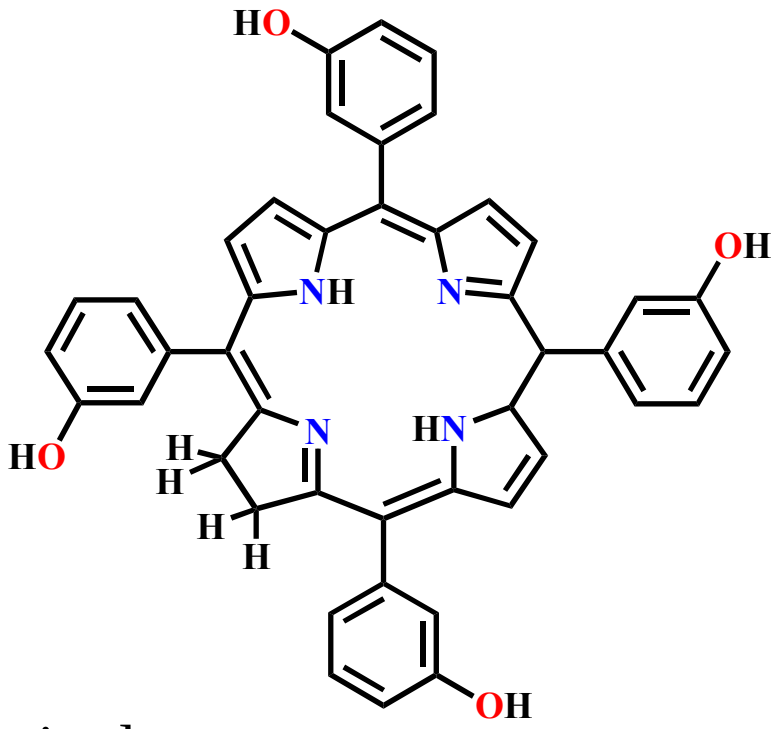
Hetacillin for ampicillin



- Ampicillin is chemically unstable in solution due to the a-NH_2 group attacking the b-lactase ring
- 'N' in heteracillin is locked up within a heterocyclic ring

5.8 Prodrugs activated by external influences -sleeping agents

Sleeping agent. This is an inactive compound which is only converted to the active drug by some form of external influence.



Example: Photodynamic therapy - Foscan

- Inactive and accumulates in cells
- Activated by light - method of targeting tumour cells
- Foscan is excited and reacts with oxygen to produce toxic singlet oxygen
- Singlet oxygen can then attack proteins and unsaturated lipids in the cell membrane leading to the formation of hydroxyl radicals which further react with DNA leading to cell destruction

6- Drug alliances - synergism

Definition:

Drugs which have a benefical effect on the activity or pharmacokinetic properties of another drug

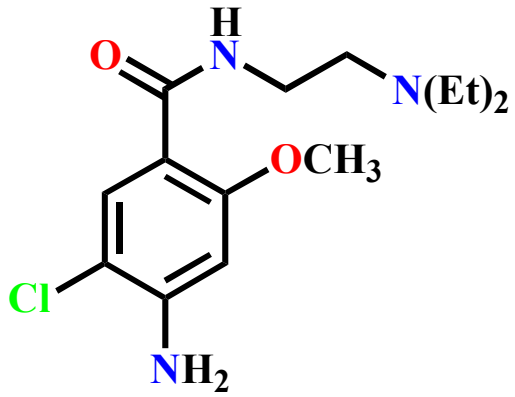
A drug causes localising drugs to a target area

Example: Adrenaline and procaine (local anaesthetic)

- Adrenaline constricts blood vessels at the injection area
- Procaine is localised at the injection area

A drug causes increasing absorption

Example: Metoclopramide



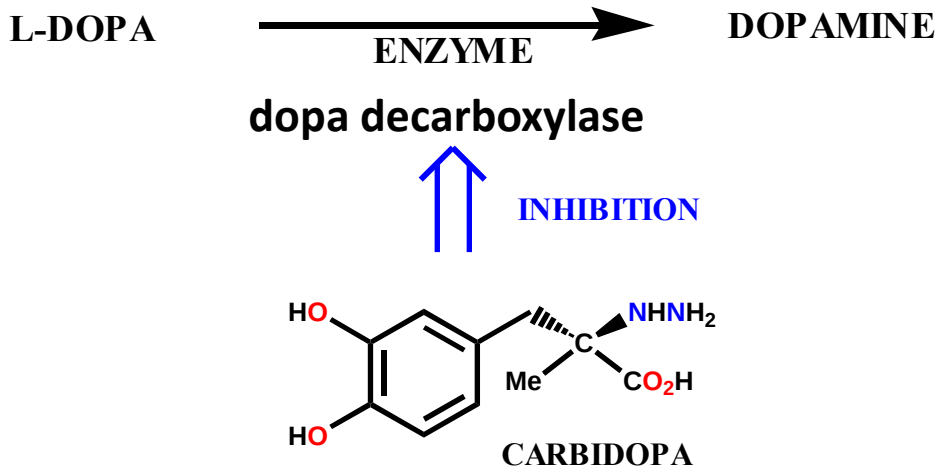
- Administered with analgesics in the treatment of migraine
- Increases gastric motility and causes faster absorption of analgesics
- Leads to faster pain relief

Sentry Drugs

A drug that is added to ‘protect’ another drug

Example:

Carbidopa



- Carbidopa protects L-dopa
- It inhibits the decarboxylase enzyme in the peripheral blood supply
- It is polar and does not cross the blood brain barrier
- It has no effect on the decarboxylation of L-Dopa in the CNS
- Smaller doses of L-dopa can be administered - less side effects

END OF LECTURE 2